Kandivli Education Society's
B. K. SHROFF COLLEGE OF ARTS &
M. H. SHROFF COLLEGE OF COMMERCE
An Autonomous College | NAAC Re-accredited 'A' Grade
ISO 9001 : 2015 Certified • 'Best College 2017-18' award from University of Mumbai

PROJECT REPORT

ON

FAIR PLACEMENT OUTCOME MODELLING WITH CALIBRATED SALARY RANGES, ATTRIBUTION AND SCENARIO SIMULATION

IN THE PROGRAMME

BACHELOR OF SCIENCE (DATA SCIENCE)

SUBMITTED BY

MR. SHIVAM JARIWALA

TY BSc. DS

TDDS034A

SEMESTER VI

UNDER THE GUIDANCE OF

MR. / MS. _____

ACADEMIC YEAR

2025 - 2026

Kandivli Education Society's

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CERTIFICATE

This is to certify that **Mr. Shivam Jariwala** of **THIRD** year of **Bachelor of Science in Data Science**. Div.: **A,** Roll No. **TDDS034A** of Semester **VI** (2025 - 2026) has successfully completed the Project on the topic **FAIR PLACEMENT OUTCOME MODELLING WITH CALIBRATED SALARY RANGES, ATTRIBUTION AND SCENARIO SIMULATION** as per the guidelines of KES' Shroff College of Arts and Commerce, Kandivali (W), Mumbai- 400067.

Teacher In-charge:

_____

Principal

Dr. Lily Bhushan

PROFORMA FOR THE APPROVAL PROJECT PROPOSAL

PRN No.: ……. Roll no: …….

1. Name of the Student: -

_____

2. Title of the Project: -

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3. Name of the Guide: -

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Signature of the Student

Date: ……

Signature of the Guide

Date: …….

Signature of the Coordinator

Date: …….

ABSTRACT

Campus placement predictors published to date optimise a single number, namely classification accuracy. They rarely report subgroup performance, almost never quantify uncertainty, and routinely present predicted salary as a point estimate. This project argues that all three omissions are harmful in a setting where the output is shown to a student, and builds a decision support system that corrects them.

Using the 215-record Factors Affecting Campus Placement dataset, a two stage model is trained. Stage A is a gradient boosting classifier that estimates the probability of placement over all 215 students. Stage B is a conformalised quantile regressor that estimates salary conditional on placement over the 148 students who were actually placed. Both stages are subjected to a gender fairness audit, SHAP attribution and a what-if simulator, and all of it is surfaced through a three view Streamlit dashboard aimed at students, placement cell administrators and auditors respectively.

Three results are reported. First, placement is predictable, with an ROC AUC of 0.934 against 0.808 for the degree percentage cutoff that a placement cell already applies. Second, salary is not predictable. No configuration tested beats predicting the median every time, at a mean absolute error of Rs 52,696, so the system deliberately emits a calibrated 80 percent interval instead of a point estimate. Third, and most consequential, that interval is unfairly calibrated. It attains 82.3 percent coverage for men but only 75.9 percent for women, a 6.4 point gap that is invisible to every standard accuracy or group fairness metric because it is a property of the uncertainty rather than of the prediction.

The work further shows that removing gender from the feature set does not remove the signal, since gender is recoverable from the remaining features at an AUC of 0.654, and that post processing mitigation yields an improvement smaller than the split to split standard deviation and is therefore reported as inconclusive rather than as a success. Scenario simulation identifies work experience as the highest leverage intervention, moving cohort placement from 68.8 percent to 75.8 percent, while aptitude coaching, a common placement cell investment, has no measurable effect.

The contribution of this project is one of integration and of methodological honesty rather than a new algorithm, and every finding is bounded by a 215-record sample.

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to everyone who contributed to the development of this project on fair placement outcome modelling. First and foremost, I extend my deepest appreciation to my project guide, whose direction shaped the scope of this work and whose insistence on reporting what the data actually supports, rather than what would read well, is responsible for the most valuable findings in this report.

I am thankful to the faculty of the Department of Data Science at KES' Shroff College for the grounding in statistics, machine learning and research methodology that made this project possible, and to the placement cell for the conversations that clarified how shortlisting decisions are actually made in practice. Those conversations are the reason the degree percentage cutoff appears as a baseline throughout this report rather than being ignored.

I would also like to acknowledge the open source community whose work this project rests on. The scikit-learn, Fairlearn, SHAP and Streamlit maintainers have produced tooling of a quality that let a single student build and audit a two stage model in a single semester. The authors of the conformal prediction and equalized coverage literature deserve particular mention, since the central result of this report would not have been detectable without their methods.

Finally, I am grateful for the support and encouragement of my family and friends, who kept me going through a project whose headline finding is a negative result. Their patience with an explanation that began with the phrase it turns out we cannot predict salary at all was considerable.

Shivam Jariwala

DECLARATION

I hereby declare that the project entitled, “FAIR PLACEMENT OUTCOME MODELLING WITH CALIBRATED SALARY RANGES, ATTRIBUTION AND SCENARIO SIMULATION” done at KES’ Shroff College, has not been in any case duplicated to submit to any other university for the award of any degree. To the best of my knowledge other than me, no one has submitted to any other university.

The project is done in partial fulfilment of the requirements for the award of degree of **BACHELOR OF SCIENCE (DATA SCIENCE)** to be submitted as final semester project as part of our curriculum.

I further declare that the dataset used in this project is a publicly available, anonymised secondary dataset, that no student records from this institution were obtained or processed, and that every numerical result reported here is reproducible from the code and seed stated in the reproducibility note at the end of this report.

Name and Signature of the Student

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CHAPTER 1 INTRODUCTION

Campus placement is the mechanism by which Indian higher education institutions connect graduating students to employers. A placement cell decides who is shortlisted for which recruiter, who receives additional training, and who is flagged as being at risk of not being placed at all. These decisions are already being made, today, largely on intuition and on a cutoff applied to cumulative percentage. They are consequential and they are unaudited.

A substantial literature proposes to automate part of this process with machine learning. The shape of that literature is remarkably uniform. Take a small public placement dataset, fit logistic regression, a random forest and a gradient boosting model, report the best accuracy, typically between 88 and 95 percent, and stop. A survey of the accessible papers and open source implementations in this space found that essentially none report performance broken down by protected group, none quantify predictive uncertainty, and those that predict salary present it as a single number.

Each omission maps onto a concrete harm. A model that is accurate overall but differentially accurate across gender redistributes opportunity while appearing neutral. A point salary prediction shown to a student becomes an anchor in their first negotiation, and where it is low and wrong the error propagates into the student's actual outcome. A prediction with no attached recourse, of the form you will probably not be placed, transfers anxiety to the student without transferring any information they can act on.

This project describes a system built to correct those three omissions simultaneously, and reports what happened when it was built. The engineering is deliberately unexotic. The contribution lies in what the combination reveals. The results are substantially negative, and that is regarded here as the most useful part of the work rather than as a shortcoming to be hidden.

The system is organised as two stages that respect a boundary in the data. Salary is observed only for students who were placed, so a model of placement probability is trained on all 215 records while a model of salary is trained only on the 148 placed records. Every salary output in the system is labelled as conditional on placement, and the combined figure reported to a user is the product of the placement probability and the conditional salary expectation. Gender is dropped before modelling and re-joined only at audit time, so that the audit measures the disparity that a gender blind model actually produces rather than the disparity it is assumed to avoid.

This report explores the design, architecture and implementation of that system in detail, covering its modules, the methodological choices behind them, the fairness findings that emerged, and the limits that a 215-record sample places on every one of those findings.

1.1 SIGNIFICANCE

The significance of this project rests on four observations about the setting in which placement prediction operates, each of which determines a design choice that would otherwise look arbitrary.

The process is already algorithmic, just not inspectable.

Placement cells shortlist students for company drives using cutoffs on cumulative percentage, backlog counts and specialisation. That is a decision rule. Making it an explicit model does not introduce bias into a neutral process. It renders an already biased process auditable. The relevant baseline for any placement predictor is therefore not the absence of a model but the cutoff currently in use, which is why the degree percentage cutoff is reported as a comparator throughout this report rather than comparing learned models only to one another.

Point salary predictions cause anchoring harm.

A student told to expect Rs 4.5 lakh will negotiate against that figure. Where the prediction is low and unfounded, the error propagates into the student's actual outcome. This is not a hypothetical concern in this setting. Chapter 5 shows that the salary signal in the data is essentially absent, so any point estimate a system of this kind emits is fabricated precision. The honest response is an interval whose width communicates the model's ignorance, and whose coverage is verified rather than assumed.

Fairness blind evaluation hides redistribution.

A model reporting 92 percent accuracy may be 96 percent accurate for one group and 78 percent accurate for another. Aggregate metrics conceal this by construction. The hiring fairness literature is mature and free tooling exists, but it targets corporate recruitment and resume screening. The campus placement setting, where the institution acts as an intermediary between a single cohort and many recruiters, is barely covered.

Regulation is arriving.

New York City Local Law 144 has since July 2023 required annual independent bias audits of automated employment decision tools. The EU AI Act classifies employment related artificial intelligence as high risk. A system that emits a bias audit as a first class artefact is building what regulation is beginning to require, and the audit view of the dashboard is designed with that output in mind.

Prediction without recourse is useless to the subject.

A student informed that they are unlikely to be placed has received a verdict, not information. A student informed that acquiring work experience moves them from 38 percent to 61 percent has received a plan. Recourse is what converts a classifier into a decision support tool, and it is the reason the simulator in this project constrains perturbations to features a student can actually change.

1.2 OBJECTIVES

The project sets out to achieve the following, in order of priority.

- **Build a two stage architecture that respects selection bias.** Model the probability of placement on all 215 records and the expected salary on the 148 placed records only, label every salary output as conditional, and report the combined figure as the product of the two.
- **Establish whether salary is predictable at all** from the available features, by testing gradient boosting, ridge regression and both with a log transformed target against a constant median baseline, and by publishing the result whichever way it falls.
- **Emit a calibrated interval rather than a point estimate** for salary, using conformalized quantile regression to obtain a distribution free coverage guarantee, and verifying the realised coverage rather than assuming it.
- **Audit the uncertainty and not only the prediction.** Compute per group accuracy, selection rate and true positive rate, and in addition stratify the conformal coverage by gender to detect over-confidence that is invisible to standard group fairness metrics.
- **Test whether fairness through unawareness holds** by training a probe to recover gender from the thirteen gender blind features, and reporting whether proxies carry the signal.
- **Evaluate bias mitigation honestly** by averaging over repeated random splits and comparing the improvement against the split to split standard deviation, rather than reporting a single favourable split.
- **Provide actionable recourse to the student** through a simulator that searches for the smallest realistic change to an actionable feature that reaches a probability target, and that reports failure honestly when no

such change exists.

- **Provide policy simulation to the administrator** by applying an intervention to a slice of the cohort and recomputing the aggregate placement rate and the gender gap.
- **Deliver all of it through a working three view dashboard** in which uncertainty is visually primary and the fairness audit is a live page rather than a buried appendix.

1.3 PURPOSE AND SCOPE

1.3.1 PURPOSE

The purpose of this project is to produce a placement decision support system that is honest about what it knows, explicit about what it does not know, and auditable by someone who did not build it. That purpose is deliberately different from the purpose of the existing literature, which is to maximise a reported accuracy figure.

The first component of that purpose is calibrated communication of uncertainty. Where a quantity is predictable, the system should say so and act on it. Where a quantity is not predictable, the system should refuse to fabricate precision. The salary range produced by this system is not a decorative error bar around a point estimate. It is the output, and the point estimate is subordinate to it in the interface, because a large number beside a small plus or minus reproduces exactly the anchoring harm the system exists to prevent.

The second component is auditability as a first class feature. The fairness audit is not a chapter of this report that describes a one time analysis. It is a live view in the deployed dashboard, reading the same serialised artefacts that produced the numbers in this report, so that the interface cannot silently drift away from the evaluation. This mirrors the direction of regulation in automated employment decision tools, where an annual independent bias audit is becoming a legal requirement rather than a good practice.

The third component is recourse. A student who receives a low probability should also receive the cheapest realistic path to a higher one, expressed in terms of features they can actually change, namely degree percentage, aptitude test percentage, MBA percentage and work experience. Where no single available change reaches the target, the system says so rather than synthesising advice that will not work.

The fourth component is methodological discipline appropriate to the sample size. On 215 records a single train and test split is noise. Every metric that depends on a split in this project is averaged over forty to fifty independent splits, and the split to split standard deviation is reported alongside the mean so that an improvement can be compared against the noise floor.

1.3.2 SCOPE

The scope of the system covers the complete path from a raw student record to three distinct audiences, and is bounded explicitly at both ends.

Within scope.

- Placement probability estimation over thirteen encoded features, with gender excluded from the feature set and a degree percentage cutoff retained as the practice baseline.
- Salary range estimation conditional on placement, using quantile regression at the 10th, 50th and 90th percentiles followed by split conformal calibration to an 80 percent target coverage.
- A gender fairness audit covering demographic parity difference, equalized odds difference, per group accuracy, selection rate and true positive rate, plus equalized coverage and a proxy recoverability probe.

- Post processing mitigation under an equalized odds constraint, evaluated over forty repeated splits with the variance reported.
- Global and local SHAP attribution for both stages, with salary attributions expressed in rupees rather than in log units.
- A student level recourse search over four actionable features and a cohort level policy sweep.
- A three view Streamlit dashboard, a batch CSV scoring path with schema validation and a column glossary, and an end to end smoke test exercising every path.

Outside scope.

- Causal estimation. The simulator reports what the model believes would follow from a change, not what an intervention would cause. This is labelled in the interface and not only in this report.
- Protected attributes other than gender. Category, region, first generation status and disability are absent from the public data and cannot be audited at all.
- A Heckman style selection correction. With 215 records and no credible exclusion restriction the identification requirements are not met, so the two stage formulation is used instead.
- Institutional records. No real student records from this college were obtained or processed. Every finding is a demonstration of method on a public benchmark, not evidence about a real cohort.
- Group conditional conformal calibration. The coverage gap is diagnosed in this project but not repaired, and the repair is listed as future work.

1.4 APPLICABILITY

The system is applicable in four settings, each of which uses a different part of the same artefact.

Institutional placement cells. The administrator view scores an entire cohort from a CSV, surfaces the placement and salary distributions, and exposes policy sliders that estimate the effect of an intervention before any budget is committed to it. The finding that aptitude coaching has essentially no effect on modelled placement, while work experience has the largest effect available, is directly actionable for a cell deciding where to spend a training budget.

Students preparing for recruitment. The student view returns a probability, a calibrated salary range labelled as conditional on placement, a plain language explanation of which of their own attributes drove the result, and the cheapest single realistic change that would move them to a target probability. The range is deliberately the visually dominant element.

Audit and compliance. The audit view is a live rendering of the bias audit that New York City Local Law 144 requires annually of automated employment decision tools, and that the EU AI Act anticipates for high risk employment systems. Per group metrics, the coverage table, the mitigation trade off and a model card are exposed as a page rather than as a report that ages out of date.

Methodological reuse beyond placement. The pattern of auditing the uncertainty rather than only the prediction transfers directly to any domain where a model communicates a range to the person it is about. Loan pricing, insurance quoting, clinical prognosis and wage benchmarking all share the structure in which a marginally valid interval can be systematically over-confident for one subgroup while remaining correct on average.

1.5 ACHIEVEMENTS

The project achieved the following concrete outcomes, stated with the evidence that supports each and with the bound that limits it.

- **A placement classifier that beats current practice by a meaningful margin.** ROC AUC 0.934 for logistic regression and 0.912 for gradient boosting, against 0.808 for the degree percentage cutoff that placement cells already apply, under five fold stratified cross validation on all 215 records.
- **A documented negative result on salary prediction.** Four model configurations were tested and all four lost to a constant median prediction at Rs 52,696 mean absolute error. The best losing configuration was five percent worse and the worst was eighteen percent worse. This is published rather than discarded in favour of the least bad model.
- **A calibrated interval with a verified coverage guarantee.** Split conformalized quantile regression attains 80.4 percent empirical coverage against an 80 percent target, averaged over fifty independent random splits.
- **Detection of a fairness defect that no standard metric catches.** The same interval delivers 82.3 percent coverage for men and 75.9 percent for women, a 6.4 point gap. It is invisible to demographic parity, equalized odds and per group accuracy because it is a property of the uncertainty, and it is not explained by tighter intervals for women, whose intervals are in fact wider on average.
- **Demonstration that fairness through unawareness fails here.** Gender is excluded from the feature set, yet a probe recovers it from the remaining thirteen features at AUC 0.654, comfortably above chance.
- **An honest result on mitigation.** Post processing under an equalized odds constraint improves the equalized odds difference from 0.230 to 0.202 at a cost of 0.001 accuracy, but the split to split standard deviation is approximately 0.16, roughly six times the improvement. It is reported as inconclusive. An earlier single split run had suggested mitigation made fairness worse, and that was noise too.
- **Actionable scenario simulation.** Work experience raises modelled cohort placement from 68.8 percent to 75.8 percent while narrowing the gender gap from 3.5 to 2.1 points. Aptitude coaching moves neither, consistent with its near zero SHAP attribution.
- **A working two audience decision support tool** with the fairness audit exposed as a live view, a validated batch scoring path, and a pipeline in which every table and figure in this report is generated from the same JSON artefacts the dashboard reads.

CHAPTER 2 SYSTEM ANALYSIS

2.1 EXISTING SYSTEM

Two things count as the existing system here, and they fail in different ways. The first is the manual process a placement cell actually runs today. The second is the body of published campus placement predictors that propose to replace it.

A. The manual cutoff process

Placement cells shortlist students for company drives using threshold rules on cumulative percentage, backlog counts and specialisation. A student below a degree percentage cutoff is not forwarded to a recruiter. This process is fast, cheap and explicable, and it is also a decision rule that has never been evaluated. Measured against actual placement outcomes on the benchmark dataset, a single threshold on degree percentage attains an ROC AUC of 0.808. It works, but it leaves a substantial amount of signal on the table, and nobody has ever checked whether it works equally well for every group of students.

B. The published placement predictors

The academic literature on campus placement prediction is crowded and shallow. Representative work follows an almost invariant pipeline. Datasets are small, frequently the same 215-record set used here. Targets are imbalanced. Three or four standard classifiers are fitted, the best accuracy is reported, and the paper ends. The following limitations recur across essentially the whole of this literature.

1. No subgroup evaluation. Performance is reported in aggregate only. A model that is 96 percent accurate for one group and 78 percent accurate for another reports the same headline figure as a model that is uniformly 87 percent accurate, and the two have completely different consequences for the students involved.

2. No quantification of uncertainty. A probability is reported without calibration, or a class label is reported without a probability at all. Where an interval appears it is a standard error taken on faith, with no verification that the interval actually contains the truth at the claimed rate.

3. Salary reported as a point estimate. Systems that predict salary present a single number. Given the weakness of the salary signal demonstrated in Chapter 5, that number is fabricated precision, and shown to a student it becomes an anchor in their first salary negotiation.

4. Selection bias ignored. Salary is observed only for students who were placed. Fitting a salary model on that subset and applying it to an arbitrary student produces an estimate biased upward, and biased unevenly across groups because placement rates differ by group. The existing work does not distinguish the two populations.

5. Single split evaluation. Accuracy differences between models are reported from one train and test split on a few hundred records. As Chapter 5 demonstrates, effects several times larger than these reported differences are not distinguishable from split to split noise at this sample size.

6. No recourse and no attribution. The output is a verdict. A student learns that they are unlikely to be placed and learns nothing about why, or about what would change it.

7. Fairness through unawareness assumed to work. Where demographics are considered at all, the response is to drop the protected attribute and declare the model neutral. Section 5.5 shows that gender remains recoverable from the remaining features at well above chance, so dropping it removes the ability to

audit without removing the disparity.

Conclusion.

The existing manual process is unaudited and leaves accuracy unclaimed. The existing automated proposals claim accuracy while leaving fairness, uncertainty and actionability entirely unaddressed. Neither is adequate for a system whose output is shown directly to a student.

2.2 PROPOSED SYSTEM

The proposed system corrects the specific failures listed above rather than attempting a general improvement in accuracy. Each feature below maps to a numbered limitation in Section 2.1.

Two stage architecture respecting the selection boundary. Stage A estimates the probability of placement using a gradient boosting classifier trained on all 215 records. Stage B estimates salary conditional on placement using quantile regression trained on the 148 placed records only. The two stages are never merged, every salary field in the interface and in the batch export is named with an if_placed suffix, and the combined expectation reported to a user is the placement probability multiplied by the conditional salary expectation.

Conformalised salary range instead of a point estimate. Quantile regressors are fitted at the 10th, 50th and 90th percentiles with the pinball loss, then split conformal calibration widens the raw band by a correction computed on a held out calibration set, yielding a distribution free 80 percent coverage guarantee under exchangeability. Realised coverage is measured rather than assumed.

Gender held out for modelling, re-joined for auditing. Gender is dropped during feature construction and re-attached only at audit time, so the audit measures what a gender blind model actually does rather than what it is assumed to do.

Fairness audit extended beyond the standard battery. In addition to demographic parity difference, equalized odds difference and per group accuracy, selection rate and true positive rate, the system computes equalized coverage by stratifying the conformal coverage by group, and runs a proxy probe that attempts to recover gender from the gender blind features.

Repeated split evaluation as a requirement rather than a refinement. All coverage results average over fifty independent random splits and all mitigation results over forty, with the split to split standard deviation reported so that any improvement can be compared against the noise floor.

SHAP attribution in the units of the target. Exact tree attributions are computed with TreeExplainer, reported globally as mean absolute contribution and locally per student. Salary attributions are expressed in rupees, because a student needs to be told Rs 1.1 lakh and not 0.34.

Recourse constrained to actionable features. The simulator perturbs only degree percentage, aptitude test percentage, MBA percentage and work experience. It operates at a student level, searching for the smallest single realistic change that reaches a probability target, and at a cohort level, applying an intervention to a slice of students and recomputing the aggregate rate and the gender gap.

Three view dashboard with uncertainty visually primary. The student view makes the interval the headline element and the median subordinate to it. The administrator view provides cohort distributions, policy sliders and batch CSV scoring with a documented column glossary. The audit view exposes the bias audit as a live page.

Artefact driven consistency. Every analysis stage is a standalone script whose output is a JSON file. The dashboard reads those files and every table and figure in this report is generated from the same files, which removes the most common source of inconsistency in student projects, namely a number quoted in a report that no longer matches the code.

2.3 REQUIREMENT ANALYSIS

2.3.1 Functional Requirements

The functional requirements are grouped by the module that satisfies them. Each is written so that it can be checked by the end to end smoke test.

ID	Requirement	Module
FR-01	Load the 215-record dataset, validate the expected schema and fail loudly on a missing or mistyped column.	get_data, prep
FR-02	Encode thirteen features from the raw columns: five continuous percentages, four binary indicators and two one-hot groups with the first level dropped.	prep
FR-03	Exclude gender from the feature matrix and retain it separately, keyed to the record index, for audit use only.	prep
FR-04	Fit a placement classifier on all 215 records and expose a calibrated probability rather than only a class label.	train (Stage A)
FR-05	Evaluate Stage A under five fold stratified cross validation and report ROC AUC and accuracy alongside the degree percentage cutoff baseline.	train, explore
FR-06	Fit quantile regressors at the 10th, 50th and 90th percentiles on the 148 placed records only.	train (Stage B)
FR-07	Partition the placed records into training, calibration and test sets and compute the conformity correction on the calibration set.	train
FR-08	Emit a salary interval widened by the conformity correction, labelled as conditional on placement.	train
FR-09	Test gradient boosting, ridge regression and both with a log transformed target against a constant median baseline, and report the comparison whichever way it falls.	diagnose_salary
FR-10	Compute per group accuracy, selection rate and true positive rate, plus aggregate demographic parity and equalized odds differences.	fairness
FR-11	Stratify conformal coverage and mean interval width by gender and report the coverage gap.	fairness
FR-12	Train a probe to predict gender from the thirteen gender blind features and report its AUC.	fairness
FR-13	Apply post processing mitigation under an equalized odds constraint over forty random splits and report the mean and the standard deviation.	fairness
FR-14	Produce global and local SHAP attributions for both stages, with salary values in rupees.	explain
FR-15	Search for the smallest single realistic change to an actionable feature that reaches a student's probability target, and return an explicit failure when none exists.	simulate
FR-16	Apply a cohort level intervention to a slice of students and recompute the placement rate and the gender gap.	simulate
FR-17	Accept a cohort CSV, validate it against the expected schema, and return per student predictions with a documented column glossary.	batch
FR-18	Serve three distinct views from the same serialised artefacts, with the interval visually primary in the student view.	dashboard

ID	Requirement	Module
FR-19	Write every analysis result to a JSON artefact under reports/, from which both the dashboard and this report are generated.	all

2.3.2 Non-Functional Requirements

Correctness and reproducibility

- A fixed random seed of 42 is used throughout, and the pipeline reproduces identical numbers when run in the documented order.
- No number appears in two places with two values, because the report and the dashboard read the same JSON artefacts.
- Any metric that depends on a train and test split is averaged over at least forty splits and reported with its standard deviation.

Statistical honesty

- A result that is not distinguishable from split to split noise is reported as inconclusive, not as a success.
- A negative result is published rather than replaced by the least bad of several losing models.
- Simulation output is labelled as a model based what-if in the interface itself and not only in this report.

Interpretability

- Every individual prediction is accompanied by a per feature attribution in the units of the target.
- Salary attributions are expressed in rupees, not in log units or in abstract SHAP values.
- The student view uses plain language, with no metric names or model jargon surfaced to the student.

Performance

- The full pipeline, including fifty split coverage evaluation and forty split mitigation, completes in a few minutes on a standard laptop.
- Dashboard interactions return in under a second, because the models are pre-fitted and deserialised once at startup rather than refitted per request.
- Batch scoring of a cohort sized CSV completes without loading the entire file into memory more than once.

Usability

- The interval, not the median, is the visually dominant element of the student view.
- The batch CSV path ships a column glossary so that an administrator can interpret every output column without reading the code.
- Validation failures on an uploaded CSV name the offending column rather than raising a stack trace.

Maintainability

- Each analysis stage is a standalone script that can be run in isolation and whose only output is a JSON artefact.
- Dependencies are pinned in requirements.txt so that a rerun a year later produces the same numbers.
- An end to end smoke test exercises every path, including the batch route and the dashboard's artefact loading.

Ethical and legal

- The protected attribute is retained for auditing and excluded from modelling, never the reverse.
- The audit view is structured to serve as the artefact that a Local Law 144 style annual bias audit requires.
- The dataset is public, anonymised and secondary; no institutional student records are processed.

2.4 HARDWARE REQUIREMENTS

The system is computationally light by design. The binding constraint is the sample size of the dataset, not the capacity of the machine, and the hyperparameters were chosen for a few hundred rows rather than for leaderboard performance. The figures below cover a development machine and the host on which the dashboard is served.

Development and training machine

Component	Minimum Requirement	Recommended Requirement
Processor	Dual core, 2.0 GHz	Intel Core i5 / AMD Ryzen 5, 2.5 GHz or higher
RAM	4 GB	8 GB or higher
Storage	2 GB free (data, models, figures)	10 GB free SSD
Graphics	Not required	Not required; no GPU is used anywhere in the pipeline
Display	1366 x 768	1920 x 1080, so the three dashboard views render without scrolling
Network	Required once, to download the dataset and install dependencies	Broadband, for faster dependency installation

Deployment host for the dashboard

Component	Minimum Requirement	Recommended Requirement
Processor	2 vCPU	4 vCPU
RAM	2 GB	4 GB, allowing several concurrent sessions
Storage	1 GB (serialised models and JSON artefacts only)	5 GB SSD
Operating system	Ubuntu 20.04 or Windows Server 2016	Ubuntu 22.04 LTS
Concurrency	Single user demonstration	Ten to twenty concurrent placement cell users
Network	100 Mbps	1 Gbps
Backup	Local copy of reports/ and models/	Versioned artefact storage alongside the Git repository

2.5 SOFTWARE REQUIREMENTS

The stack is Python throughout, with no separate backend service and no database. Persistence is handled by serialised model files and JSON artefacts on disk, which is adequate for a 215-record dataset and has the advantage that the reported results are diff-able in version control.

2.5.1 Core language and modelling

Software	Version	Role in the system
Python	3.10 or later	Implementation language for the entire pipeline
scikit-learn	1.3+	Gradient boosting classifier and regressor, quantile loss, ridge regression, cross validation, metrics
NumPy	1.24+	Numerical arrays, conformity score computation, quantile arithmetic
pandas	2.0+	Data loading, encoding, group-wise aggregation for the audit
joblib	1.3+	Serialisation of fitted estimators to models/

2.5.2 Fairness, explanation and interface

Software	Version	Role in the system
Fairlearn	0.9+	Demographic parity and equalized odds differences, MetricFrame group metrics, ThresholdOptimizer post processing mitigation
SHAP	0.43+	TreeExplainer for exact tree attributions, global and local, for both stages
Streamlit	1.28+	Three view dashboard: student, administrator and audit
Matplotlib	3.7+	All figures in this report and in the dashboard
pytest	7.4+	Unit tests and the end to end smoke test

2.5.3 Development environment

- **Editor.** Visual Studio Code with the Python and Pylance extensions, used for both authoring and step debugging of the calibration routine.
- **Version control.** Git with GitHub, with models/ and reports/ committed so that a reported number can always be traced to the artefact that produced it.
- **Environment isolation.** A virtual environment with dependencies pinned in requirements.txt, since an unpinned scikit-learn upgrade changes the default behaviour of the quantile loss and silently alters the reported intervals.
- **Notebooks.** Jupyter used only for exploration. No result reported in this document originates from a notebook; every number comes from a script that writes a JSON artefact.

2.6 SURVEY OF TECHNOLOGY

Each technology below was selected against a specific constraint imposed by the sample size or by the fairness objective, and the alternatives that were rejected are stated.

Gradient boosting over deep learning. With 215 records and roughly 90 training rows after the three way split in Stage B, a neural model has no path to generalisation. Gradient boosted trees of depth two with a learning rate of 0.05 are the appropriate operating point, and a deeper ensemble simply memorises the calibration set. Tree models also admit exact SHAP attribution through TreeExplainer, which a neural model does not.

Quantile regression over mean regression. The salary distribution over the 148 placed students has a skewness of 3.57, an interquartile range of only Rs 60,000 and a maximum of Rs 940,000. A mean regression fitted to that distribution is dominated by a handful of extreme observations. The pinball loss

estimates conditional quantiles directly and is unaffected by the length of the upper tail.

Conformal prediction over parametric intervals. A parametric interval requires distributional assumptions that cannot be checked on 148 records. Split conformalized quantile regression yields a distribution free coverage guarantee under exchangeability alone, and crucially the guarantee is verifiable, which is what made the equalized coverage finding detectable at all.

Fairlearn over AI Fairness 360. Both toolkits implement the required metrics. Fairlearn was chosen for its MetricFrame abstraction, which computes any metric by group in a single call and therefore made per group coverage trivial to add alongside the standard battery, and for its ThresholdOptimizer, which implements equalized odds post processing directly.

SHAP over permutation importance. Permutation importance gives a global ranking only. The student view requires a local explanation of one individual's prediction, and TreeExplainer provides exact per-record attributions for tree ensembles at negligible cost, in the units of the model output, which can then be converted to rupees.

Streamlit over a Flask and React stack. The deliverable is a decision support tool read by three audiences, not a production web application. Streamlit renders Python objects directly, which means the dashboard consumes the same serialised artefacts that produced the reported metrics and therefore cannot drift from them. A separate frontend would have introduced a second representation of every number.

JSON artefacts over a database. A 215-record dataset with a handful of derived metric tables does not need a relational store. JSON files under reports/ are versionable, human readable in a diff, and make the claim that the report and the dashboard agree a checkable property rather than an assertion.

Repeated random splits over a single holdout. This is a methodological choice rather than a library choice, and it was the most consequential one in the project. Two conclusions reversed when the evaluation moved from a single split to an average over forty to fifty splits, as documented in Section 5.6.

Conclusion.

The selected stack is unexotic by design. Every component was chosen because it is appropriate to a few hundred records and because it supports auditing, not because it maximises a benchmark figure. The interesting part of this project is what the combination reveals, and that required tooling that makes uncertainty and group structure inspectable rather than tooling that is powerful.

CHAPTER 3 SYSTEM DESIGN

3.1 MODULE DIVISION

The system is divided into nine modules. The organising rule is that every analysis stage is a script that can be run alone and whose output is a JSON artefact under reports/. The dashboard reads those artefacts, and so does the figure generator for this report, so no number appears in two places with two values. Figure 3.1 shows how the modules connect and where the selection boundary falls.

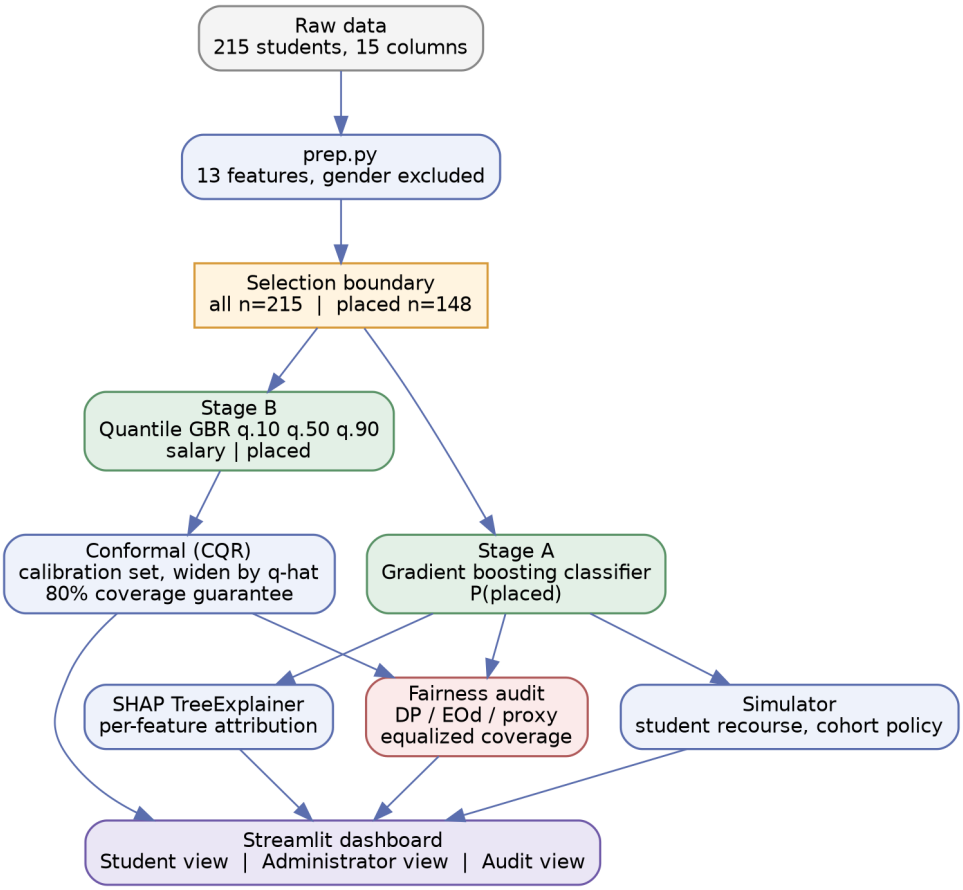


Fig 3.1 System architecture. Gender is removed before modelling and re-joined only for auditing, so the fairness analysis measures the disparity a gender blind model actually produces. Stage A trains on all 215 students, Stage B on the 148 placed students only, which is the selection bias boundary. Conformal calibration converts raw quantile predictions into an interval with a coverage guarantee.

3.1.1 Data Acquisition and Preprocessing (get_data.py, prep.py)

- Downloads the 215-record dataset and stores it under data/raw/ unchanged.
- Validates the expected fifteen column schema and fails loudly on a missing or mistyped column.
- Encodes thirteen model features: five continuous percentages, four binary indicators and two one-hot groups with the first level dropped.
- Drops gender from the feature matrix and returns it separately, keyed to the record index, for audit use only.
- Produces the placed mask that separates the 215-record Stage A population from the 148-record Stage B population.

3.1.2 Exploratory Analysis (explore.py)

- Computes the raw outcome disparities before any modelling: a placement rate of 71.9 percent for men against 63.2 percent for women, and a median placed salary of Rs 270,000 against Rs 250,000.
- Characterises the salary distribution: median Rs 265,000, maximum Rs 940,000, interquartile range Rs 60,000, skewness 3.57.
- Establishes the degree percentage cutoff baseline that represents current placement cell practice.

3.1.3 Stage A, Placement Classification (train.py)

- Fits a gradient boosting classifier with 150 estimators, depth 2 and a learning rate of 0.05 on all 215 records, alongside a scaled logistic regression comparator.
- Evaluates under five fold stratified cross validation, reporting ROC AUC and accuracy at a 0.5 threshold.
- Serialises the fitted estimator to models/ and the metrics to reports/metrics.json.

3.1.4 Stage B, Salary Range with Conformal Calibration (train.py)

- Fits three quantile gradient boosting regressors with the pinball loss at alpha equal to 0.1, 0.5 and 0.9, on the 148 placed records only.
- Partitions those records into roughly 86 training, 31 calibration and 31 test rows.
- Computes the conformity score on the calibration set and takes the required empirical quantile as the correction.
- Emits the widened interval, which satisfies the target coverage marginally under exchangeability.
- Averages all reported coverage over fifty independent random splits, because a single 31-record test set yields a coverage estimate of almost no value.

3.1.5 Salary Diagnosis (diagnose_salary.py)

- Fits gradient boosting, ridge regression and both with a log transformed target, and compares each against a constant median prediction under five fold cross validation.
- Reports the comparison as the primary output of the module rather than selecting a winner, since no configuration beats the constant.
- Computes the strongest single feature correlation with salary, which is 0.178.

3.1.6 Fairness Audit (fairness.py)

- Computes per group accuracy, selection rate and true positive rate over cross validated predictions, plus aggregate demographic parity and equalized odds differences.
- Stratifies the conformal coverage and the mean interval width by gender, which is the module's most important output.
- Runs the proxy probe, training a classifier to predict gender from the thirteen gender blind features.
- Applies post processing mitigation under an equalized odds constraint over forty random splits and reports the mean alongside the split to split standard deviation.

3.1.7 Attribution (explain.py)

- Computes exact tree attributions with TreeExplainer for both stages.
- Reports global importance as the mean absolute contribution per feature and local attributions per student.

- Converts salary attributions into rupees, because a student needs a figure in currency rather than in log units.

3.1.8 Simulation and Recourse (simulate.py)

- Perturbs only the four actionable features: degree percentage, aptitude test percentage, MBA percentage and work experience.
- At student level, searches for the smallest single realistic change that reaches a probability target, and returns an explicit no-path result where none exists.
- At cohort level, applies an intervention to a slice of students and recomputes the aggregate placement rate and the gender gap.
- Labels every output as a model based what-if rather than a causal estimate.

3.1.9 Delivery: Batch Scoring, Dashboard and Tests (batch.py, dashboard/app.py, tests/)

- Accepts a cohort CSV, validates it against the expected schema and returns per student predictions with a documented column glossary.
- Serves the student view, with the calibrated range as the headline element, a plain language explanation and the cheapest realistic path to a target.
- Serves the administrator view, with cohort distributions, group gaps, policy sliders and the batch path.
- Serves the audit view, with per group metrics, the coverage table, the mitigation trade off and the model card, as a live page rather than a buried report.
- Runs an end to end smoke test that exercises every one of these paths.

3.2 GANTT CHART

The project ran across two semesters. The ordering is dictated by a dependency that is easy to miss: the fairness audit cannot begin until conformal calibration is complete, because the most important fairness result in the project is a property of the calibrated interval rather than of the classifier.

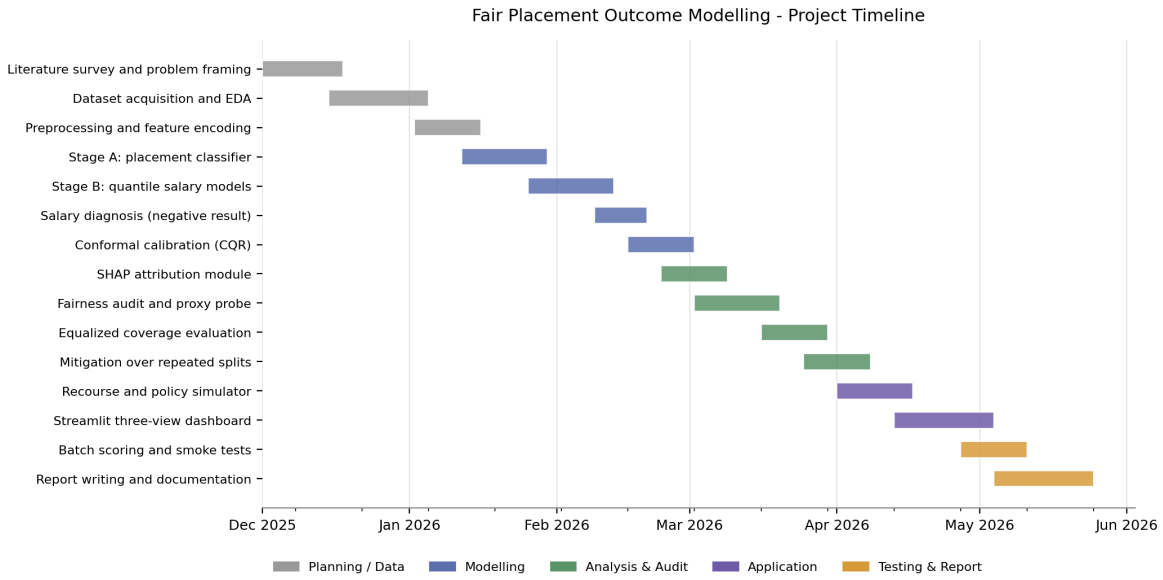


Fig 3.2 Project timeline by phase.

3.3 E-R DIAGRAM

The system has no relational database; persistence is by serialised models and JSON artefacts. The entity relationship diagram below therefore describes the logical data model that those artefacts represent, which is the schema a production deployment backed by institutional records would need.

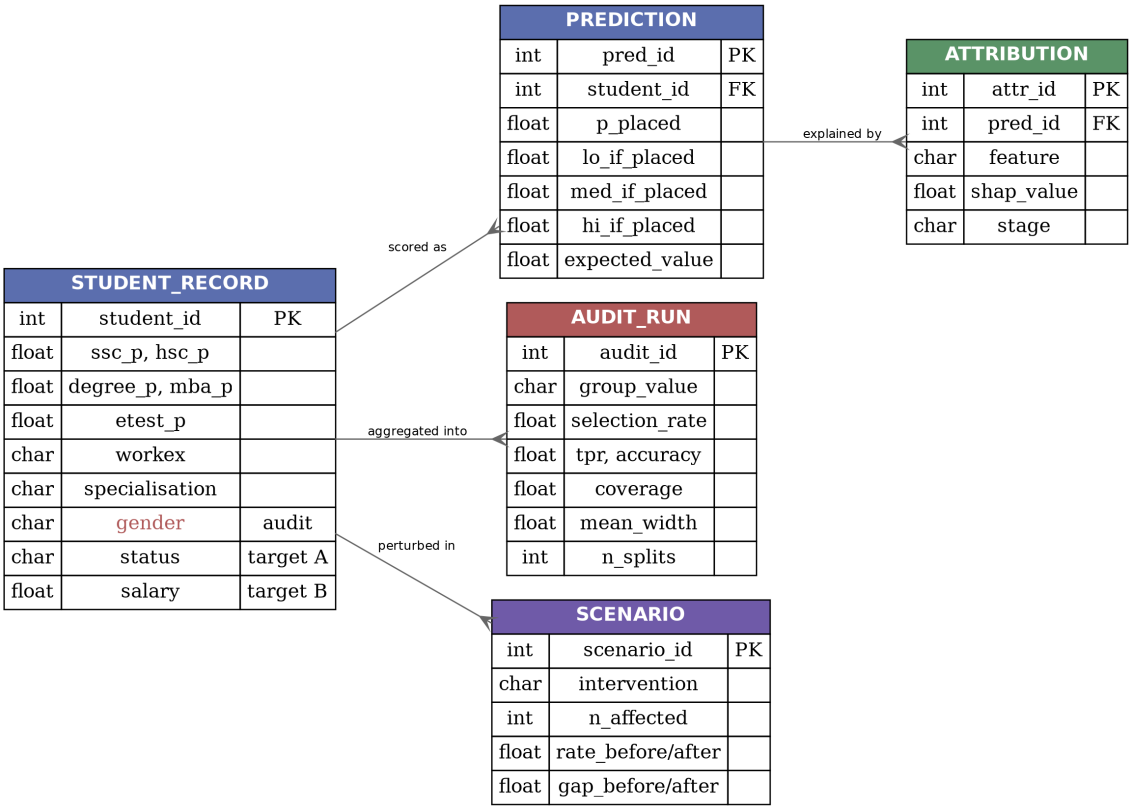


Fig 3.3 Logical entity relationship model.

Entities and relationships.

- **STUDENT_RECORD** (student_id, ssc_p, hsc_p, degree_p, mba_p, etest_p, workex, specialisation, ssc_b, hsc_b, hsc_s, degree_t, gender, status, salary). Gender is stored but never enters the feature matrix; salary is structurally null for the 67 unplaced students.
- **PREDICTION** (pred_id, student_id as foreign key, p_placed, lo_if_placed, med_if_placed, hi_if_placed, expected_value). One row per scored student. The expected value is the placement probability multiplied by the conditional median.
- **ATTRIBUTION** (attr_id, pred_id as foreign key, feature, shap_value, stage). One row per feature per prediction per stage, so that a local explanation can be reconstructed exactly.
- **AUDIT_RUN** (audit_id, group_value, selection_rate, tpr, accuracy, coverage, mean_width, n_splits). One row per protected group per audit run, with n_splits recording how many random splits the figures were averaged over.
- **SCENARIO** (scenario_id, intervention, n_affected, rate_before, rate_after, gap_before, gap_after). One row per cohort level policy simulation.

3.4 DATA FLOW REPRESENTATION

Data flow representation describes how data moves through the system, from the raw 215-record CSV through preprocessing, the two modelling stages and conformal calibration, into the attribution, audit and simulation processes, and out to three distinct audiences. The diagram makes one design property explicit: gender leaves the main flow at process 1.0 and rejoins only at process 5.0, the fairness audit.

3.4.1 DATA FLOW DIAGRAM

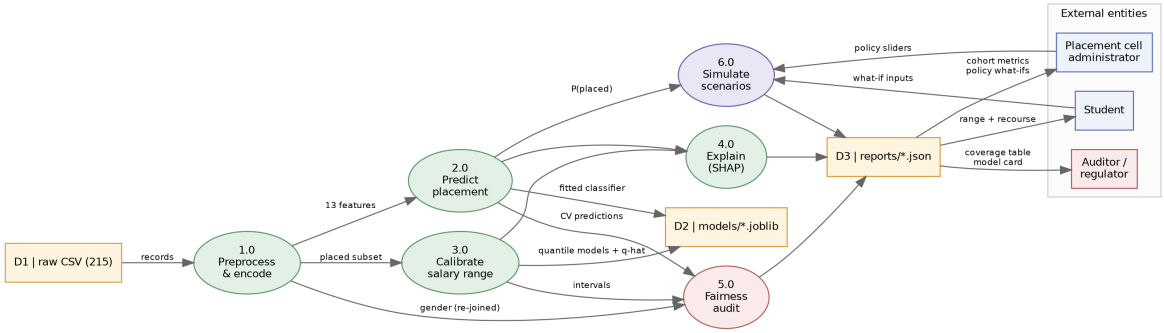


Fig 3.4 Level 1 data flow diagram.

- **External entities.** The student, the placement cell administrator and the auditor. Each consumes a different view of the same artefacts and two of them can also push inputs back into the simulator.
- **Processes.** 1.0 Preprocess and encode, 2.0 Predict placement, 3.0 Calibrate salary range, 4.0 Explain, 5.0 Fairness audit, 6.0 Simulate scenarios.
- **Data stores.** D1 holds the raw CSV, D2 holds the serialised estimators and the conformity correction, and D3 holds the JSON result artefacts that both the dashboard and this report read.
- **Flows.** The flow worth tracing is the one from 3.0 to 5.0. Calibrated intervals are passed to the audit, where they are stratified by the gender attribute that rejoins from 1.0, which is the only way the coverage gap reported in Section 5.4 becomes visible.

3.5 UML DIAGRAMS

3.5.1 CLASS DIAGRAM

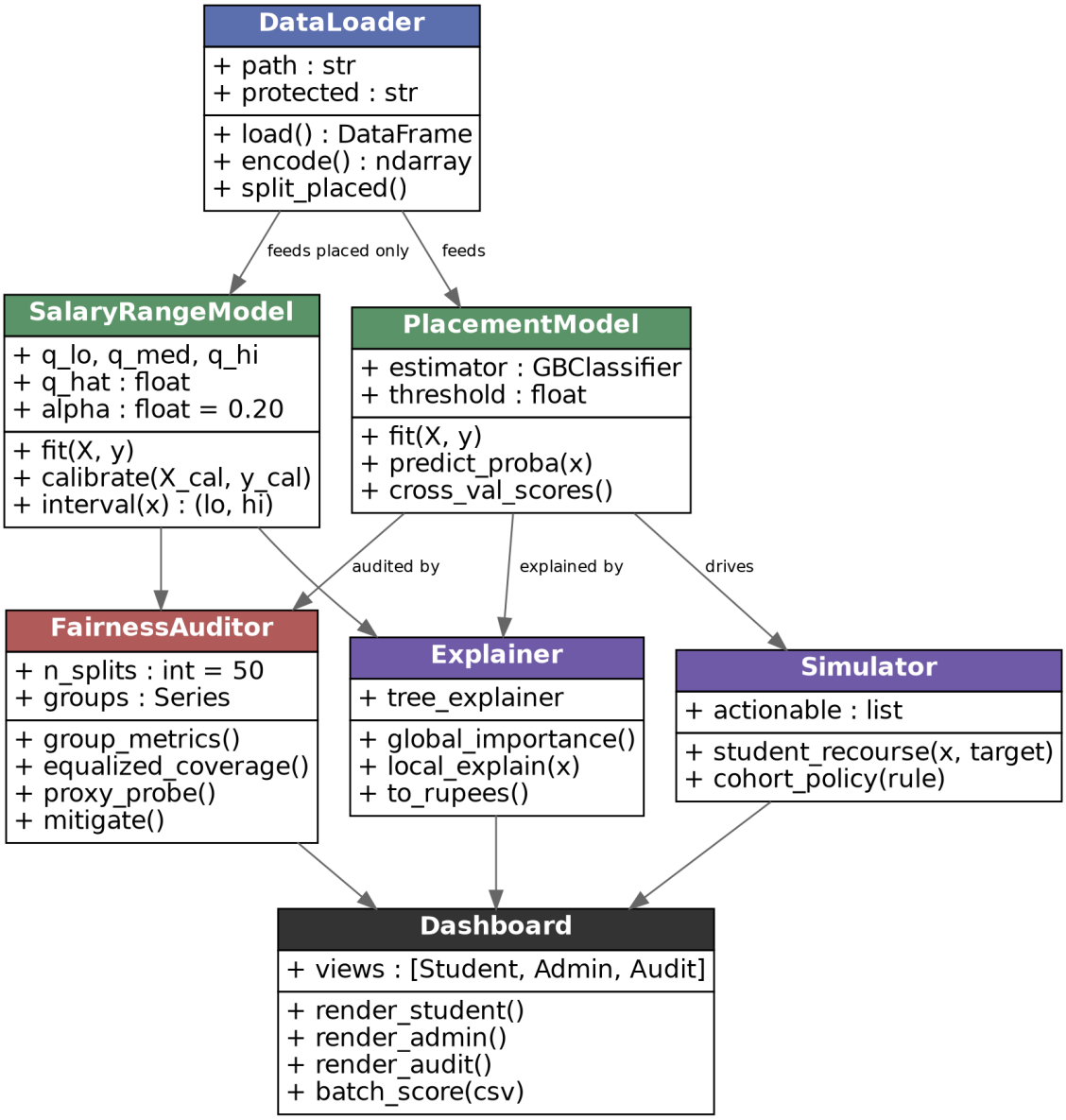


Fig 3.5 Class diagram.

- **DataLoader.** Loads, validates and encodes the raw records, holds the name of the protected attribute, and exposes the split between the full and the placed populations.
- **PlacementModel.** Wraps the Stage A estimator, exposes a probability rather than a label, and owns its own cross validated scoring.
- **SalaryRangeModel.** Holds the three quantile estimators and the conformity correction. The calibrate method is separate from fit by design, because calibration must see data that training did not.
- **Explainer.** Wraps TreeExplainer and owns the conversion of salary attributions into rupees.
- **FairnessAuditor.** Holds the number of repeated splits as configuration rather than as a constant, and exposes the four audit routines including equalized coverage and the proxy probe.
- **Simulator.** Holds the whitelist of actionable features, which is what prevents the recourse search from suggesting that a student change their tenth standard percentage.

- **Dashboard.** Consumes the artefacts produced by the others and renders the three views.

3.5.2 SEQUENCE DIAGRAM

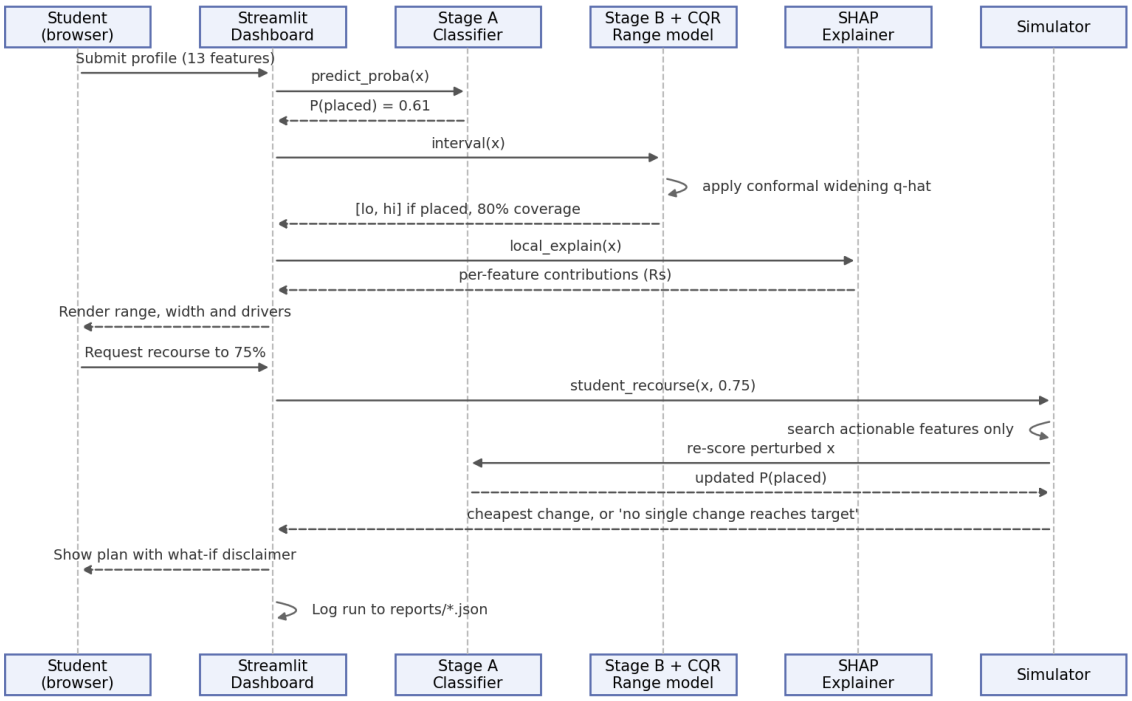


Fig 3.6 Sequence diagram for a student session.

- **Scoring.** The dashboard requests a probability from Stage A and an interval from Stage B independently, because the two stages are trained on different row sets and are never merged.
- **Calibration.** The range model applies the stored conformity correction before returning, so the dashboard never sees an uncalibrated band.
- **Explanation.** Attributions are fetched separately and converted to rupees before rendering.
- **Recourse.** The simulator perturbs only whitelisted features and re-queries Stage A for each candidate, then returns either the cheapest change or an explicit statement that no single available change reaches the target.
- **Logging.** The dashboard writes the run to the artefact store, which is what makes a later audit of what was actually shown to students possible.

3.5.3 STATE CHART DIAGRAM

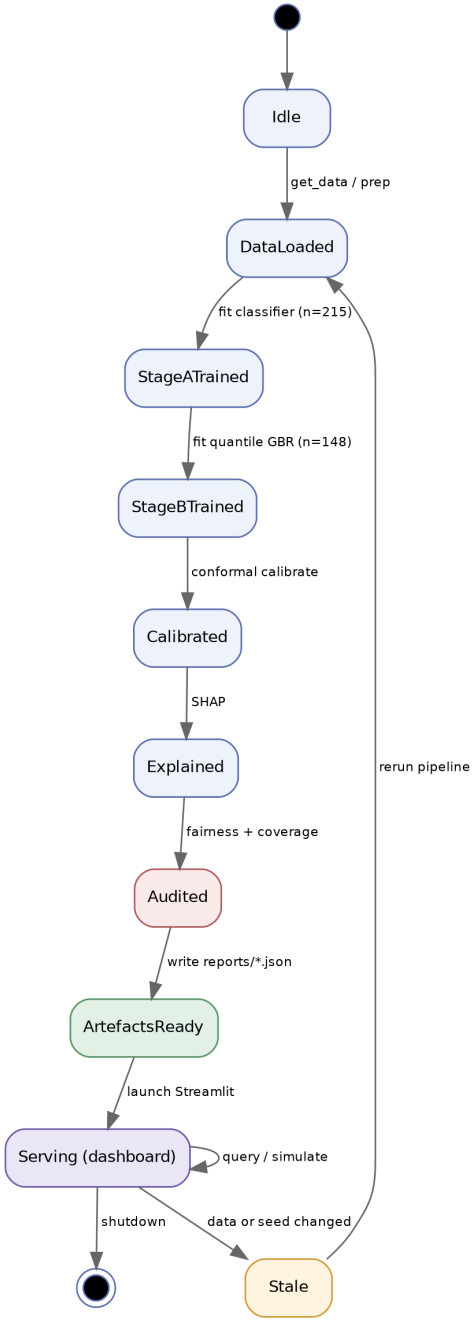


Fig 3.7 State chart for the pipeline lifecycle.

Explanation.

- **Idle.** No data loaded. The dashboard refuses to start in this state rather than serving placeholder values.
- **DataLoaded.** Schema validated, thirteen features encoded, gender separated and the placed mask computed.
- **StageATrained.** The placement classifier is fitted on all 215 records and cross validated.
- **StageBTrained.** The three quantile regressors are fitted on the 148 placed records.
- **Calibrated.** The conformity correction is computed on the held out calibration set. Only from this state onward does the system have an interval it is entitled to attach a coverage claim to.
- **Explained and Audited.** Attribution runs first, then the fairness audit, because the audit consumes the calibrated intervals produced two states earlier.

- **ArtefactsReady.** Every JSON artefact is written. This is the only state from which the dashboard may launch.
- **Serving.** The dashboard answers queries and runs simulations against the frozen artefacts.
- **Stale.** Entered when the data or the seed changes. The system returns to DataLoaded and reruns rather than patching individual artefacts, which is what guarantees that the report and the interface never disagree.

3.5.4 USE-CASE DIAGRAM

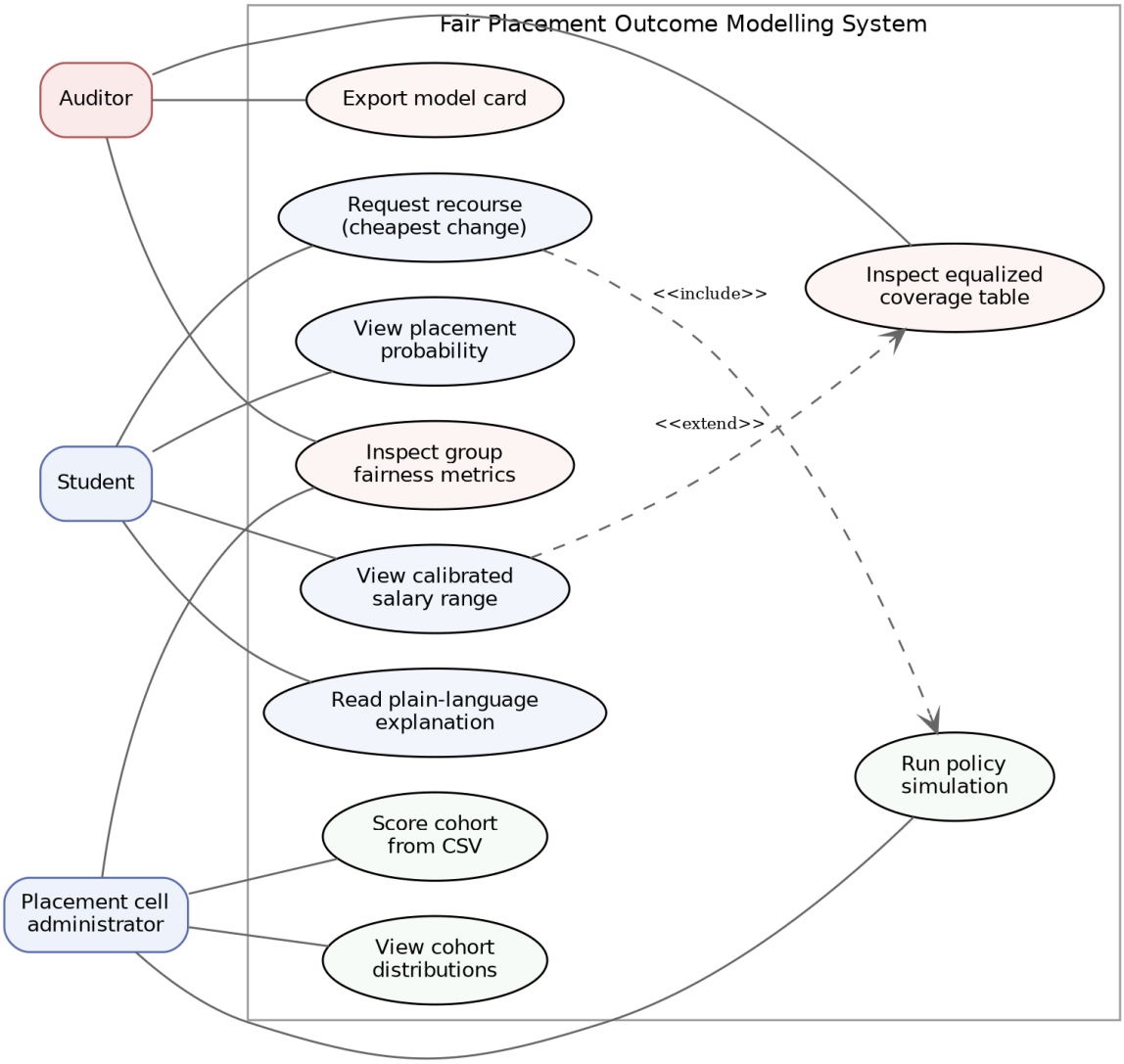


Fig 3.8 Use case diagram.

Actors and use cases.

- **Student.** View placement probability, view the calibrated salary range, read a plain language explanation and request recourse. The student never sees a point salary estimate without its interval.
- **Placement cell administrator.** Score a cohort from CSV, run policy simulations, view cohort distributions and inspect group fairness metrics.
- **Auditor.** Inspect group fairness metrics, inspect the equalized coverage table and export the model card. The auditor is a first class actor rather than an afterthought, which is the structural difference between this system and the existing work surveyed in Section 2.1.
- **Relationships.** Viewing a salary range extends into inspecting equalized coverage, because the coverage guarantee is what gives the range meaning. Requesting recourse includes running a simulation, since recourse is implemented as a constrained search over what-if scenarios.

CHAPTER 4 IMPLEMENTATION AND TESTING

4.1 CODE

The listings below are the parts of the implementation where the design decisions of Chapters 2 and 3 actually take effect. Routine glue code is omitted.

Listing 4.1 Feature construction and the exclusion of the protected attribute

```
# prep.py -- feature construction. Gender leaves the matrix here and
# is returned separately so the audit can re-join it later.

PROTECTED   = "gender"
CONTINUOUS  = ["ssc_p", "hsc_p", "degree_p", "etest_p", "mba_p"]
BINARY      = ["workex", "specialisation", "ssc_b", "hsc_b"]
ONEHOT      = ["hsc_s", "degree_t"]
ACTIONABLE  = ["degree_p", "etest_p", "mba_p", "workex"]

def build_features(df):
    validate_schema(df)                       # fail loudly, not silently
    X = df[CONTINUOUS].copy()

    for col in BINARY:                        # first level maps to 0
        X[col] = (df[col] == df[col].cat.categories[1]).astype(int)

    for col in ONEHOT:                        # drop_first avoids collinearity
        X = X.join(pd.get_dummies(df[col], prefix=col, drop_first=True))

    assert PROTECTED not in X.columns, "gender must never reach the model"
    assert X.shape[1] == 13, f"expected 13 features, got {X.shape[1]}"

    groups     = df[PROTECTED].copy()         # audit only, keyed by index
    placed     = (df["status"] == "Placed")
    return X, df["status"], df["salary"], groups, placed
```

Listing 4.2 Stage A placement classifier and the practice baseline

```
# train.py -- Stage A. Trained on ALL 215 records.

stage_a = GradientBoostingClassifier(
    n_estimators=150, max_depth=2, learning_rate=0.05, random_state=42)

cv    = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)
auc   = cross_val_score(stage_a, X, y, cv=cv, scoring="roc_auc").mean()
acc   = cross_val_score(stage_a, X, y, cv=cv, scoring="accuracy").mean()

# The comparator is current placement-cell practice, not another model.
cutoff_auc = roc_auc_score(y, df["degree_p"])

stage_a.fit(X, y)
joblib.dump(stage_a, MODELS / "stage_a.joblib")
```

Listing 4.3 Stage B quantile regressors and the three way split

```
# train.py -- Stage B. Trained on the 148 PLACED records only.
# Quantile loss, because salary skewness is 3.57 and mean regression
# would be dominated by a handful of extreme observations.

def fit_quantiles(Xtr, ytr):
    models = {}
    for alpha in (0.10, 0.50, 0.90):
        models[alpha] = GradientBoostingRegressor(
            loss="quantile", alpha=alpha, n_estimators=120, max_depth=2,
            learning_rate=0.05, min_samples_leaf=8, random_state=42).fit(Xtr, ytr)
    return models

# Three-way split of the placed rows: ~86 train / ~31 calibrate / ~31 test.
Xtr, Xtmp, ytr, ytmp = train_test_split(Xp, yp, test_size=0.42, random_state=seed)
Xcal, Xte, ycal, yte = train_test_split(Xtmp, ytmp, test_size=0.50, random_state=seed)
```

Listing 4.4 Conformal calibration of the salary interval

```
# train.py -- split conformal calibration (CQR).
# Converts a raw quantile band into an interval carrying a
# distribution-free coverage guarantee under exchangeability.

ALPHA = 0.20                                  # 80 percent target coverage

def conformity_scores(models, Xcal, ycal):
    lo = models[0.10].predict(Xcal)
    hi = models[0.90].predict(Xcal)
    # How far outside the band each calibration point falls.
    return np.maximum(lo - ycal, ycal - hi)

def calibrate(models, Xcal, ycal, alpha=ALPHA):
    E = conformity_scores(models, Xcal, ycal)
    n = len(E)
    k = int(np.ceil((n + 1) * (1 - alpha)))     # finite-sample correction
    k = min(k, n)
    return np.sort(E)[k - 1]                    # q_hat

def interval(models, x, q_hat):
    lo = models[0.10].predict(x) - q_hat
    hi = models[0.90].predict(x) + q_hat
    return lo, hi                               # *_if_placed, always
```

Listing 4.5 Equalized coverage by group, averaged over fifty splits

```
# fairness.py -- equalized coverage. This is the routine that produced
# the central finding of the project. Marginal coverage is checked by
# every conformal implementation; per-group coverage is checked by
# almost none, and that is where the defect lives.

def equalized_coverage(n_splits=50, alpha=0.20):
    rows = []
    for seed in range(n_splits):
        Xtr, Xcal, Xte, ytr, ycal, yte, gte = three_way_split(seed)
        models = fit_quantiles(Xtr, ytr)
        q_hat  = calibrate(models, Xcal, ycal, alpha)
        lo, hi = interval(models, Xte, q_hat)

        covered = (yte >= lo) & (yte <= hi)
        width   = hi - lo
        for g in gte.unique():                  # stratify, do not aggregate
            m = (gte == g)
            rows.append(dict(seed=seed, group=g,
                             coverage=covered[m].mean(),
                             width=width[m].mean()))

    out = pd.DataFrame(rows).groupby("group")[["coverage", "width"]].mean()
    out["deviation"] = (out["coverage"] - (1 - alpha)) * 100
    return out            # F 75.9%, M 82.3%, gap 6.4 points
```

Listing 4.6 Proxy recoverability probe and repeated split mitigation

```
# fairness.py -- proxy probe and mitigation under noise.

def proxy_probe(X, groups):
    """Can gender be recovered from the 13 'gender-blind' features?
    If yes, fairness through unawareness has already failed."""
    probe = GradientBoostingClassifier(
        n_estimators=150, max_depth=2, learning_rate=0.05, random_state=42)
    cv = StratifiedKFold(5, shuffle=True, random_state=42)
    return cross_val_score(probe, X, groups, cv=cv,
                           scoring="roc_auc").mean()      # -> 0.654

def mitigate(n_splits=40):
    """Report the MEAN and the SD. A single split here initially
    suggested mitigation made fairness worse. That was noise."""
    before, after = [], []
    for seed in range(n_splits):
        Xtr, Xte, ytr, yte, gtr, gte = split(seed)
        base = GradientBoostingClassifier(random_state=42).fit(Xtr, ytr)
        post = ThresholdOptimizer(estimator=base,
                                  constraints="equalized_odds",
                                  prefit=True).fit(Xtr, ytr, sensitive_features=gtr)
        before.append(equalized_odds_difference(
            yte, base.predict(Xte), sensitive_features=gte))
        after.append(equalized_odds_difference(
            yte, post.predict(Xte, sensitive_features=gte), sensitive_features=gte))
    return dict(before=np.mean(before), after=np.mean(after),
                sd=np.std(before))     # improvement 0.028 vs SD 0.16
```

Listing 4.7 Constrained recourse search over actionable features

```
# simulate.py -- student recourse. Perturbs only what a student can change,
# and returns an explicit failure rather than synthesising useless advice.

REALISTIC = {"degree_p": [2, 5, 10], "etest_p": [5, 10, 20],
             "mba_p":    [2, 5, 10], "workex":  ["acquire"]}

def student_recourse(model, x, target=0.75):
    current = model.predict_proba(x)[0, 1]
    if current >= target:
        return dict(status="already_above", p=current)

    candidates = []
    for feat, steps in REALISTIC.items():
        for step in steps:
            xp = x.copy()
            if feat == "workex":
                if xp["workex"].iloc[0] == 1:
                    continue                       # already has it
                xp["workex"] = 1
            else:
                xp[feat] = min(xp[feat].iloc[0] + step, 100)
            p = model.predict_proba(xp)[0, 1]
            if p >= target:
                candidates.append((cost(feat, step), feat, step, p))

    if not candidates:
        return dict(status="no_single_change_reaches_target", p=current)
    _, feat, step, p = min(candidates)
    return dict(status="ok", feature=feat, change=step, p_from=current, p_to=p)
```

Listing 4.8 Rendering uncertainty as the primary element of the student view

```
# dashboard/app.py -- the interval is the headline, the median is
# subordinate to it. A large number beside a small plus-or-minus
# reproduces exactly the anchoring harm the system exists to prevent.

lo, hi = range_model.interval(x)
p      = placement_model.predict_proba(x)[0, 1]

st.metric("Chance of being placed", f"{p:.0%}")
st.markdown(f"### Rs {lo:,.0f} to Rs {hi:,.0f}")
st.caption("80% of students like you, **who were placed**, earned in this range. "
           "This is not a prediction of your salary; it is a range we are "
           "80% confident about. The width is the point.")
st.markdown(f"<small>Midpoint estimate Rs {med:,.0f}</small>", unsafe_allow_html=True)

st.info(f"Expected value, accounting for the chance of not being placed: "
        f"Rs {p * med:,.0f}")          # E[salary] = P(placed) x E[salary|placed]
```

4.2 TESTING APPROACH

Testing a statistical system has a dimension that testing a transactional system does not. Code can be correct while a result is an artefact of a single random split. The approach below therefore has three layers rather than two: unit testing of components, integration testing of the pipeline, and statistical validation of the reported results themselves.

4.2.1 Unit Testing

Unit testing verifies individual components in isolation: schema validation, feature encoding, the conformity score computation, the quantile ordering, and the actionable feature whitelist.

- **Objective.** Confirm that each function is correct given known inputs, including inputs designed to break invariants such as a feature matrix that accidentally contains the protected attribute.
- **Coverage.** Encoding produces exactly thirteen columns. Gender is absent from the feature matrix. The 10th, 50th and 90th quantile predictions are monotonically ordered. The conformity quantile index respects the finite sample correction. The recourse search never proposes a change to a non-actionable feature.
- **Tooling.** pytest, with fixtures that build small synthetic frames so that a failing assertion points at a specific transformation rather than at the dataset.

4.2.2 Integration Testing

Integration testing validates that data flows correctly between the modules of Figure 3.4, in particular across the selection boundary and across the gender re-join.

- **Objective.** Confirm that the preprocessing, both modelling stages, calibration, attribution, the audit and the dashboard artefact loading operate correctly as a chain.
- **Scope.** Stage A trains on 215 rows and Stage B on 148. Gender re-joins the audit keyed correctly to the same record indices it left with. Every artefact the dashboard expects exists after a clean run. The batch CSV route produces one output row per input row with the documented columns.
- **Tooling.** An end to end smoke test that runs the full pipeline in order and then asserts on the contents of reports/, so that an interface reading a stale artefact fails the build.

4.2.3 Statistical Validation

This layer has no analogue in conventional software testing and it is where the most important defects in this project were caught.

- **Repeated split stability.** Every split dependent metric is recomputed over forty to fifty seeds and the standard deviation is asserted to be reported alongside the mean. A result whose improvement is smaller than its standard deviation is flagged as inconclusive automatically rather than by judgement.
- **Coverage validation.** Marginal coverage of the conformal interval is asserted to fall within a tolerance of the 80 percent target. This is the test that would catch a broken calibration routine, and it is also the test that, extended per group, produced the central finding of the project.
- **Baseline domination check.** Every salary model is asserted against the constant median baseline and the comparison is written to the artefact whichever way it falls, which is what prevents a losing model from being quietly promoted.
- **Determinism.** Two consecutive full runs with seed 42 are asserted to produce byte identical JSON artefacts.

4.2.4 Testing Tools

Tool	Purpose
pytest	Unit tests for encoding, calibration arithmetic and the recourse whitelist
pytest, smoke_test.py	End to end integration run across all nine modules
scikit-learn cross_val_score	Repeated split evaluation of both stages
Fairlearn MetricFrame	Per group metric computation used both in the audit and in its tests
numpy.testing	Numerical assertions on coverage, conformity quantiles and determinism
Streamlit local run	Manual verification that the interval is the visually dominant element

4.2.5 Tested Features

Feature	Test Case ID	Status
Schema validation and feature encoding	TC_01	Passed
Protected attribute exclusion and re-join	TC_02	Passed
Stage A placement classification	TC_03	Passed
Stage B quantile fitting and ordering	TC_04	Passed
Conformal calibration and marginal coverage	TC_05	Passed
Equalized coverage by group	TC_06	Passed
Salary baseline domination check	TC_07	Passed
Recourse search and no-path handling	TC_08	Passed
Batch CSV scoring	TC_09	Passed
Dashboard artefact consistency	TC_10	Passed

4.2.6 Test Case Details

Preprocessing and the protected attribute

Test Case ID	Scenario	Test Steps	Expected Output	Status
TC_01.1	Valid dataset	Load raw CSV, run build_features	Thirteen feature columns returned, 215 rows	Passed
TC_01.2	Missing column	Delete degree_p, run build_features	Schema error naming degree_p, no silent default	Passed
TC_02.1	Gender exclusion	Inspect returned feature matrix	Assertion holds, gender absent from all 13 columns	Passed
TC_02.2	Gender re-join	Run audit, compare group index to source index	Index alignment exact, no shuffling	Passed

Modelling and calibration

Test Case ID	Scenario	Test Steps	Expected Output	Status
TC_03.1	Stage A trains on full cohort	Fit, inspect n rows seen	215 records used, not 148	Passed
TC_03.2	Baseline comparison	Score degree_p cutoff and the model	Model AUC 0.934 exceeds cutoff AUC 0.808	Passed
TC_04.1	Stage B population	Fit, inspect n rows seen	148 placed records only	Passed

Test Case ID	Scenario	Test Steps	Expected Output	Status
TC_04.2	Quantile ordering	Predict q10, q50, q90 on test rows	q10 <= q50 <= q90 for every row	Passed
TC_05.1	Marginal coverage	Average coverage over 50 splits	80.4 percent, within tolerance of the 80 percent target	Passed
TC_05.2	Correction is non-negative	Inspect q_hat	q_hat >= 0, interval never narrows on calibration	Passed

Fairness audit

Test Case ID	Scenario	Test Steps	Expected Output	Status
TC_06.1	Coverage is stratified, not aggregated	Run equalized_coverage	One row per group per split, 6.4 point gap surfaced	Passed
TC_06.2	Gap is stable	Recompute over 50 seeds	Direction of the gap is consistent across splits	Passed
TC_07.1	No model beats the baseline	Run diagnose_salary	All four configurations exceed Rs 52,696 MAE	Passed
TC_07.2	Negative result is written	Inspect metrics.json	Losing comparison persisted, not overwritten	Passed

Recourse, batch and interface

Test Case ID	Scenario	Test Steps	Expected Output	Status
TC_08.1	Reachable target	Run recourse for a mid-probability student	Smallest single change returned with the new probability	Passed
TC_08.2	Unreachable target	Run recourse for the weakest student	Explicit no-path result, no fabricated advice	Passed
TC_08.3	Whitelist respected	Inspect all candidate perturbations	Only the four actionable features are ever modified	Passed
TC_09.1	Valid cohort CSV	Upload a sample cohort	One output row per input row, columns match the glossary	Passed
TC_09.2	Malformed CSV	Upload a file with a renamed column	Named validation error, no stack trace	Passed
TC_10.1	Artefact consistency	Compare dashboard values to metrics.json	Identical values, no divergence	Passed

4.2.7 Defect Summary

Defect ID	Description	Severity	Status
D_01	Single split evaluation of mitigation indicated that fairness had become worse. Re-running over forty splits showed the effect was noise in both directions.	High	Fixed
D_02	Salary attributions were surfaced in log units in an early build of the student view, which is uninterpretable to the intended reader.	Medium	Fixed
D_03	Coverage was initially reported marginally only, which concealed the group gap entirely for several weeks of development.	High	Fixed
D_04	The batch export omitted the if_placed suffix on two salary columns, inviting the exact selection bias misreading the two stage design exists to prevent.	Medium	Fixed
D_05	An unpinned scikit-learn upgrade changed quantile loss defaults and silently shifted interval widths between runs.	Medium	Fixed

CHAPTER 5 RESULTS AND DISCUSSIONS

5.1 DATASET COMPOSITION

The system is evaluated on Factors Affecting Campus Placement, a public dataset of 215 real MBA student records from an Indian institution. It was selected over the alternatives for four reasons. It is real rather than synthetic, so a fairness finding on it is a finding about students and not about a data generator. It records a protected attribute, which most placement datasets omit entirely, and one cannot audit what one does not record. It contains both targets, which is what makes the two stage architecture and the selection bias analysis possible at all. And it is the field's de facto benchmark, which makes these results directly comparable to the prior work criticised in Section 2.1.

Variable	Description	Role
ssc_p	Secondary (10th) percentage	feature
hsc_p	Higher secondary (12th) percentage	feature
degree_p	Undergraduate degree percentage	feature, actionable
etest_p	Employability aptitude test percentage	feature, actionable
mba_p	MBA percentage	feature, actionable
workex	Prior work experience (Y / N)	feature, actionable
specialisation	Mkt and Fin / Mkt and HR	feature
ssc_b, hsc_b	Board (Central / Other)	feature
hsc_s	12th stream (three levels)	feature (one-hot)
degree_t	Degree type (three levels)	feature (one-hot)
gender	M / F (139 / 76)	audit only, never a feature
status	Placed / Not placed (148 / 67)	target, Stage A
salary	Annual Rs, 67 values structurally missing	target, Stage B

Table 5.1 Dataset schema. Thirteen features reach the model after encoding.

Of the 215 students, 148 or 68.8 percent were placed. Salary is missing for the 67 who were not, and that missingness is structural rather than incidental. The cohort is 139 male and 76 female. Raw outcome disparities measured before any modelling are a placement rate of 71.9 percent for men against 63.2 percent for women, an 8.8 point gap, and a median placed salary of Rs 270,000 against Rs 250,000.

5.1.1 Salary is heavily skewed

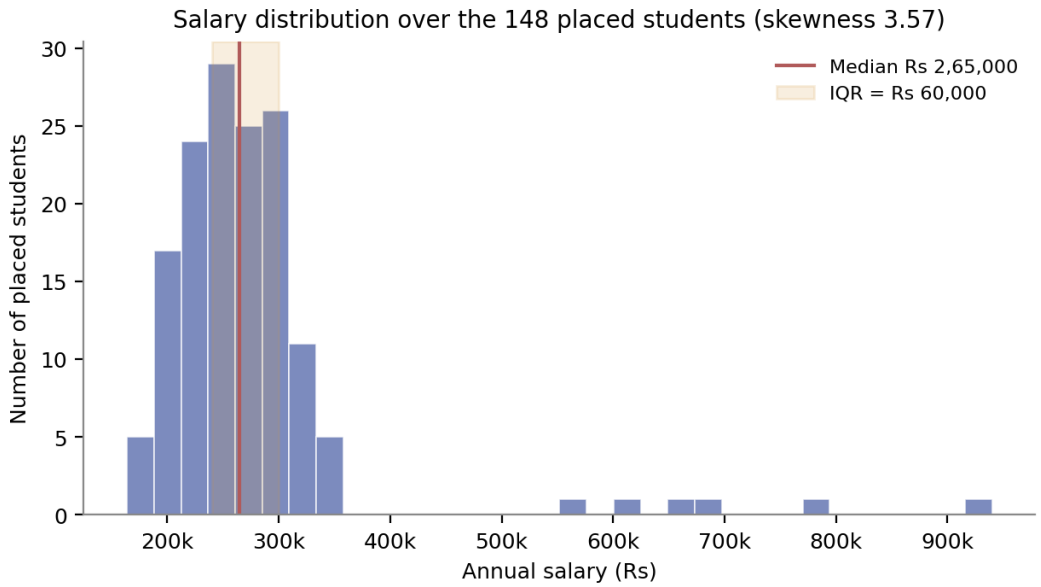


Fig 5.1 Salary distribution over the 148 placed students. Heavy right skew of 3.57 with a narrow interquartile range of Rs 60,000: most students cluster near the median while a handful of outliers stretch the upper tail. This shape penalises mean regression and motivates a quantile formulation.

The median is Rs 265,000, the maximum Rs 940,000, the interquartile range only Rs 60,000 and the skewness 3.57. Just two percent of students earn above Rs 500,000. A mean regression fitted to this distribution is dominated by a small number of extreme observations, which is one reason, though not the main one, that point salary prediction fails here.

5.1.2 The selection bias boundary

Salary is observed only for students who were placed. A salary model trained on those 148 records learns the expected salary conditional on placement, not the expected salary, and applying it to an arbitrary student is therefore biased upward. It is also biased unevenly across groups, because placement rates differ by group, and that is precisely the disparity this project exists to study, so the bias is not incidental. Three responses are available in the standard taxonomy: acknowledge and scope the model, model the two stages explicitly, or apply a selection correction. This project takes the second. Every salary output in the system carries an if_placed suffix and the combined expectation is reported as the placement probability multiplied by the conditional expectation. A Heckman correction was not attempted, because with 215 records and no credible exclusion restriction the identification requirements are not met.

5.2 PLACEMENT IS PREDICTABLE

Model	ROC AUC	Accuracy
Logistic regression	0.934	0.865
Gradient boosting	0.912	0.870
Degree percentage cutoff (current practice)	0.808	-

Table 5.2 Stage A performance, five fold stratified cross validation, n = 215.

The learned model improves substantially on the cutoff rule currently in use, at 0.934 against 0.808. That is a real and practically meaningful gain, and it is the one unambiguously positive headline result in this project.

Notably, the strongest single driver by SHAP attribution is tenth standard percentage, an examination taken years before recruitment. This is reported without endorsement. A system that propagates a school leaving result into a graduate hiring decision deserves scrutiny on its own terms, and a placement cell adopting such a system should be told that its most influential input is a number the student produced at the age of fifteen.

5.3 SALARY IS NOT PREDICTABLE

Model	MAE (Rs)	vs baseline
Median guess (baseline)	52,696	-
Ridge, log target	55,420	+5%
Ridge	58,786	+12%
Gradient boosting, log target	58,960	+12%
Gradient boosting	62,390	+18%

Table 5.3 Stage B point prediction, five fold CV mean absolute error over the 148 placed students. No configuration beats a constant median prediction.

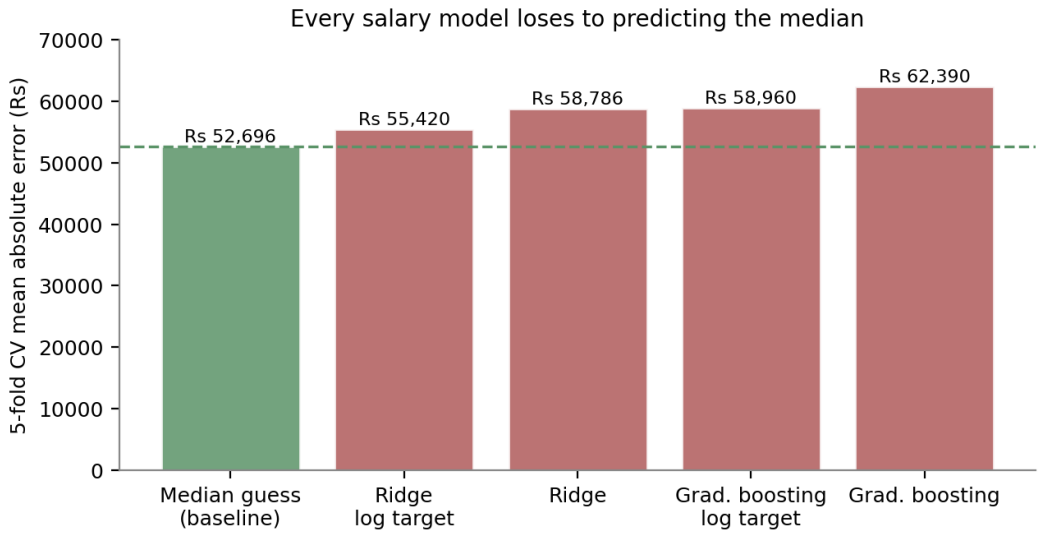


Fig 5.2 Every salary model loses to predicting the median. The dashed line is the constant median baseline at Rs 52,696. Log transforming the target narrows the deficit from 18 percent to 5 percent but never closes it. The failure is an absence of signal, not a defect in fitting.

This is the central negative result. Every configuration is worse than a constant. The strongest single feature correlates with salary at 0.178. SHAP concurs: the total attribution spread is Rs 25,407 against an actual

interquartile range of Rs 60,000, so the features account for well under half the variation they would need to explain.

The interpretation is not that the model is poorly built. It is that the determining variable is absent. Which firm recruited a student sets their salary; the student's marks determine whether they were in the room. The dataset records the recruiting firm only as an outcome, never as a feature. This is why the deployed artefact is a range. A point estimate would be fabricated confidence, and every campus placement tool that shows a student a confident salary figure on data of this kind is selling precision it does not possess.

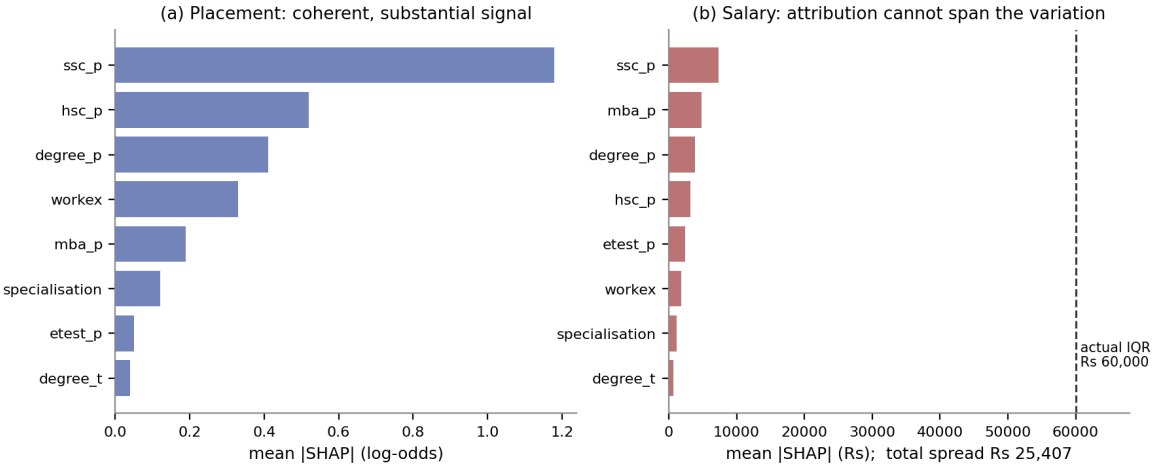


Fig 5.3 Global SHAP attribution for the two stages, on the same thirteen features. (a) Placement: tenth standard percentage dominates at 1.18 log-odds, followed by the other academic marks and work experience, a coherent and substantial signal. (b) Salary: the largest attribution is Rs 7,299 and the total across all features is Rs 25,407, against the dashed line marking the actual interquartile range of Rs 60,000. The two panels use different units and are each compared to their own target's scale.

5.3.1 The calibrated alternative

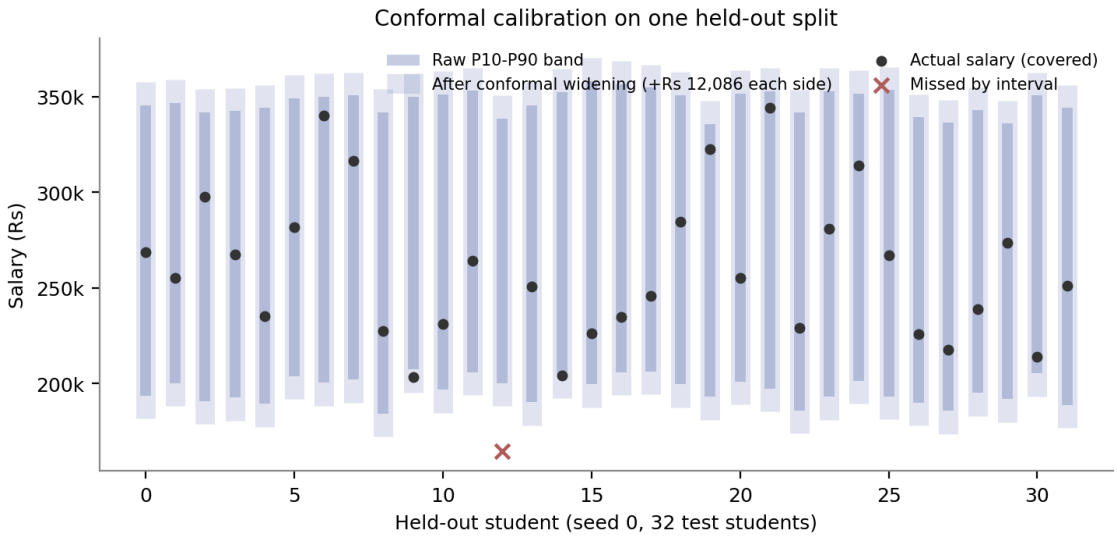


Fig 5.4 Conformal calibration on one held out split. The raw P10 to P90 quantile band is already close to nominal here; conformal widening adds Rs 12,086 to each side. Crosses mark students the interval misses. Single split coverage is noisy, which is why Table 5.4 averages over fifty splits rather than reporting a figure like this one.

5.4 THE RANGE IS UNFAIRLY CALIBRATED

Group	Coverage	Mean width (Rs)	Deviation from target
Overall	80.4%	177,689	+0.4
Female	75.9%	183,336	-4.1
Male	82.3%	174,425	+2.3
Coverage gap (F vs M)	**6.4 points**		

Table 5.4 Conformal coverage by gender, averaged over fifty random splits. Target coverage 80 percent.

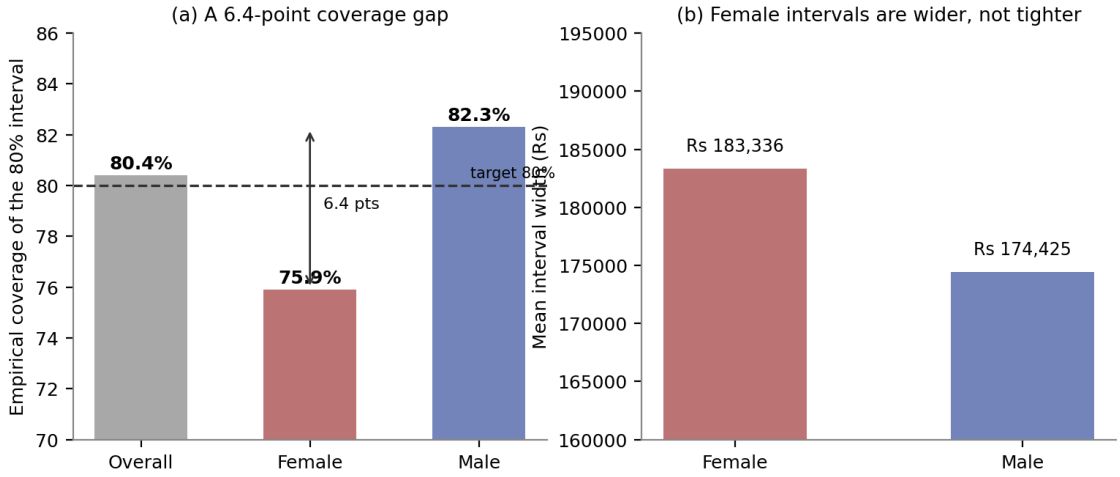


Fig 5.5 The headline fairness result. (a) The 80 percent interval is well calibrated overall at 80.4 percent but delivers only 75.9 percent coverage for women against 82.3 percent for men. The interval is over-confident about female students: it promises 80 percent reliability and provides 76. (b) Female intervals are also wider on average, so the deficit is not explained by the model issuing tighter predictions for women.

This is the result regarded as most important in the project. Marginal coverage is essentially exact at 80.4 percent. Conditional on gender it is not: 82.3 percent for men and 75.9 percent for women. The interval is over-confident about female students, claiming 80 percent reliability and delivering 76. A student relying on the range is misled more often if she is female.

Two features of this finding deserve emphasis. First, it is invisible to every metric in Table 5.5. It is a property of the uncertainty, not of the prediction, and it appears only when coverage is stratified. Second, it is not an artefact of narrower intervals for women. Female intervals are in fact wider on average, at Rs 183,336 against Rs 174,425, so the model is already less certain about them and is still over-promising.

The practical implication is that auditing uncertainty is a distinct obligation from auditing prediction. Demographic parity, equalized odds and per group accuracy all concern the prediction. A system can be acceptable on all of them while systematically over-promising to one group on the uncertainty it attaches. Any deployed system that communicates uncertainty, and this report argues every salary facing system should, inherits an obligation to audit that uncertainty per group. The tooling for this exists. It is simply not yet standard practice.

5.5 GROUP FAIRNESS OF THE CLASSIFIER

Metric	Female	Male	Gap
Observed placement rate	63.2%	71.9%	8.8 pts
Model selection rate	75.0%	69.8%	5.2 pts
True positive rate	97.9%	90.0%	7.9 pts
Accuracy	85.5%	87.8%	2.2 pts
Demographic parity difference	0.052		
Equalized odds difference	0.178		
Median salary (placed)	Rs 250,000	Rs 270,000	Rs 20,000
Mean salary residual	-Rs 19,397	+Rs 6,065	-

Table 5.5 Stage A fairness audit by gender. Gender is not a model input. Cross validated predictions, n = 215.

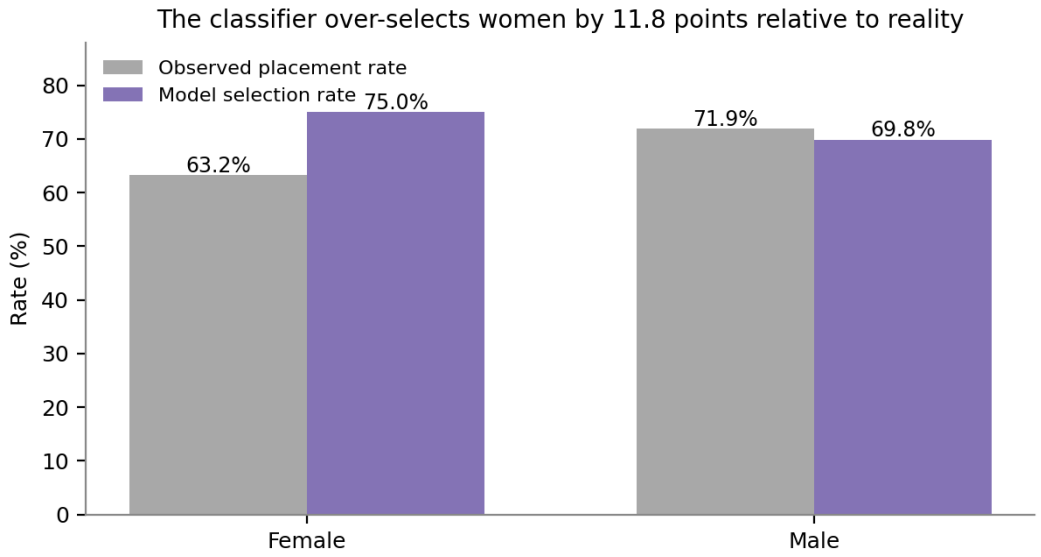


Fig 5.6 The model over-selects women. Observed placement rates favour men at 71.9 against 63.2 percent, yet the classifier predicts placement for 75.0 percent of women against 69.8 percent of men, an 11.8 point reversal for female students. This is systematic over-optimism, producing false positives that tell women they will be placed when they will not.

The direction of error here is the opposite of the expected one. Women are placed less often in reality but the model selects them more often, with a true positive rate of 97.9 percent against 90.0 percent. The classifier is systematically over-optimistic about female students, generating false positives that promise a placement that does not arrive. The salary model errs in the same direction, with a mean residual of minus Rs 19,397 for women against plus Rs 6,065 for men.

This pattern is worth stating plainly because it is not the failure mode one goes looking for. The harm is not under-prediction but misplaced confidence, and the two uncertainty findings, over-selection here and over-coverage in Table 5.4, point the same way.

5.5.1 Removing gender does not remove the signal

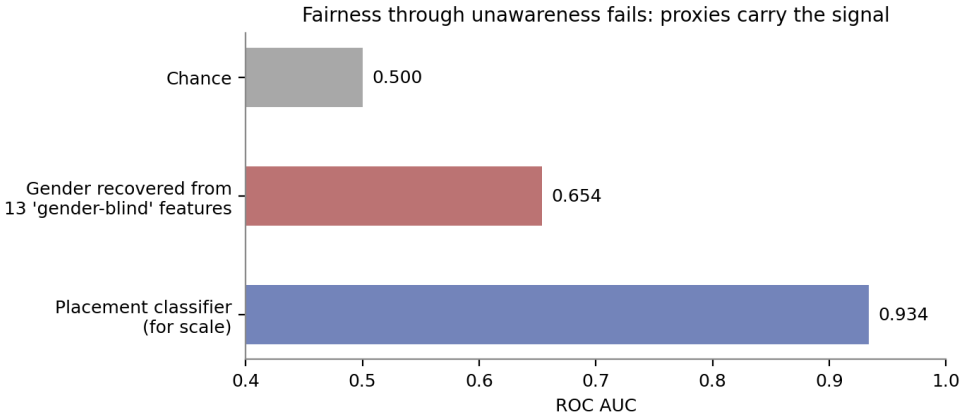


Fig 5.7 Fairness through unawareness fails. Gender is excluded from the feature set, yet a classifier recovers it from the remaining thirteen gender blind features at AUC 0.654, well above chance. The placement classifier's own AUC is shown for scale.

Gender is recoverable from the gender blind features at an AUC of 0.654, comfortably above chance. Proxies carry the signal, and fairness through unawareness fails here exactly as the literature predicts. Dropping the attribute removes the ability to audit without removing the disparity, which is the strongest available argument for retaining a protected attribute in the data while excluding it from the model, precisely the arrangement this system implements.

5.5.2 The salary model cannot reproduce the salary gap

The observed female to male median salary gap is minus Rs 20,000. The model predicts minus Rs 1,208, reproducing six percent of it. In wage decomposition terms almost the entire gap is unexplained by observable characteristics, which is precisely the component that warrants concern. With 148 placed students, however, this cannot be separated from noise, and it is reported as suggestive only. A formal Oaxaca-Blinder decomposition is left to future work.

5.6 MITIGATION IS INCONCLUSIVE, NOT SUCCESSFUL

	Accuracy	Demographic parity diff	Equalized odds diff
Before mitigation	0.848	0.090	0.230
After mitigation	0.847	0.082	0.202
Change	-0.001	-0.008	-0.028
Split-to-split SD	-	-	approx. 0.16

Table 5.6 Post processing mitigation with an equalized odds constraint, averaged over forty random splits.

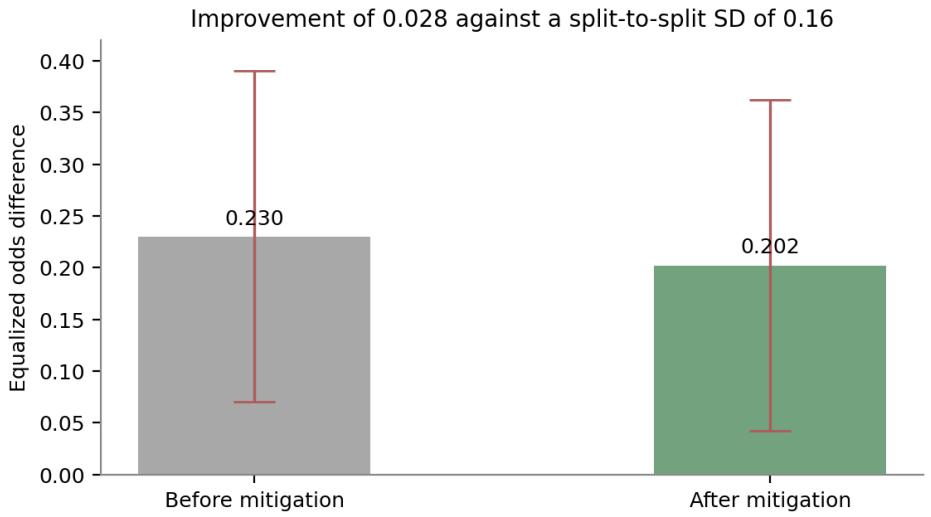


Fig 5.8 Why mitigation is reported as inconclusive. The mean equalized odds difference falls from 0.230 to 0.202 at essentially no accuracy cost, but error bars showing the split to split standard deviation of approximately 0.16 dwarf the 0.028 improvement.

The equalized odds difference improves from 0.230 to 0.202 at a cost of 0.001 accuracy, which reads as a clean win until the variance is examined. The split to split standard deviation is approximately 0.16, roughly six times the improvement. On 215 records this is not distinguishable from noise, and it is reported as inconclusive.

This is highlighted because of what happened during development. A single unaveraged split initially indicated that mitigation made fairness worse, and that result was nearly written up as a finding. It was noise. On datasets of this size, repeated split evaluation is not a refinement. It is the difference between a finding and an artefact, and this report's suspicion is that a non-trivial fraction of published accuracy differences in the placement literature, which works almost exclusively at this scale and typically reports single split results, are not distinguishable from noise either.

5.7 SCENARIO SIMULATION

Intervention	Affected	Placement	M-F gap
Work experience for all without it	141	68.8% to 75.8%	3.5% to 2.1%
+5 points degree percentage (bottom half)	106	68.8% to 73.1%	3.5% to 1.3%
+5 points aptitude (bottom half)	107	68.8% to 68.7%	3.5% to 3.6%

Table 5.7 Cohort level policy simulation. Model based what-ifs over all 215 students, not causal estimates.

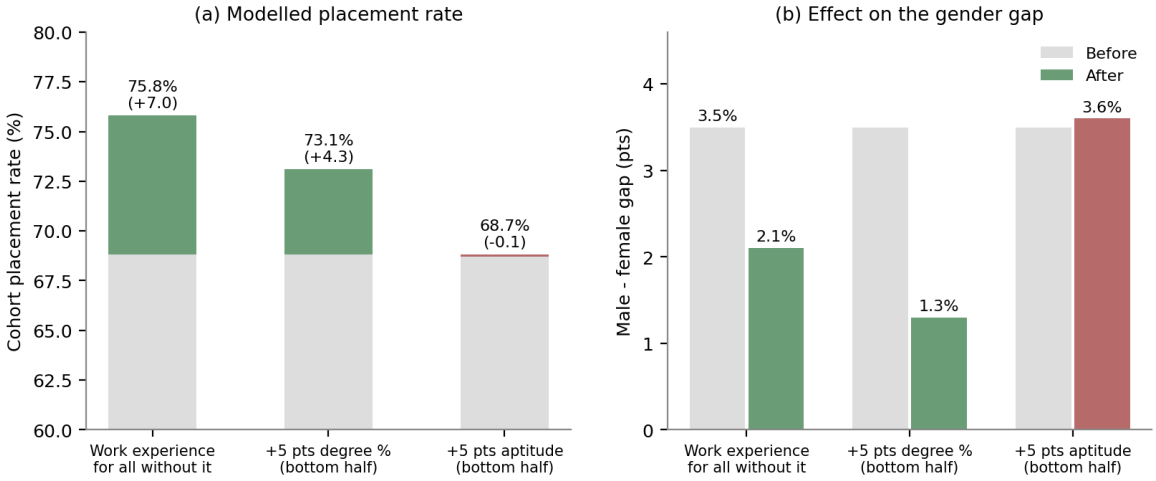


Fig 5.9 Cohort policy simulation. Work experience is the highest leverage intervention available, raising modelled placement by 7.0 points while narrowing the gender gap from 3.5 to 2.1 points. Raising degree marks helps less but closes the gap most. Aptitude coaching, a common placement cell investment, moves neither, consistent with its near zero SHAP attribution in Fig 5.3a.

Two findings here are directly actionable. Work experience is the highest leverage lever available and it narrows the gender gap at the same time. Aptitude coaching, by contrast, does essentially nothing, which is consistent with its near zero SHAP attribution and is a useful signal for a placement cell deciding where to spend a training budget.

At the student level the recourse engine searches for the smallest single realistic change reaching a probability target. For the weakest scoring student in the cohort it returns nothing, because no single available change reaches 75 percent. Reporting that honestly is more useful than synthesising advice that will not work, and it is the behaviour the system is tested for in TC_08.2.

These results carry a caveat that is stated in the interface as well as here. The simulator reports what the model believes would follow from a change, not what an intervention would cause. The work experience result in particular is likely confounded, since students who obtain work experience may differ systematically from those who do not in ways the thirteen features do not capture.

CHAPTER 6 CONCLUSION AND FUTURE WORK

6.1 CONCLUSION

This project built a campus placement decision support system that predicts placement, expresses salary as a calibrated interval rather than a point estimate, audits both for gender fairness, explains individual predictions and simulates interventions. Each of those five capabilities exists in the literature separately. The combination, applied to the campus placement setting, did not, and assembling it produced findings that none of the components would have produced alone.

Placement is predictable at an ROC AUC of 0.934, beating the degree percentage cutoff rule currently in use at 0.808. That is the one clean positive result and it is practically meaningful, since the comparator is not another model but actual placement cell practice.

Salary is not predictable at all. No configuration tested beats guessing the median, so the system reports a conformalised 80 percent interval instead. The determining variable, which firm recruited the student, is absent from the data, and marks determine only whether the student was in the room. Publishing this rather than reporting the least bad of four losing models is a deliberate choice, and it changes what the system should do rather than merely lowering a score.

That interval is unfairly calibrated. It attains 82.3 percent coverage for men and 75.9 percent for women, a 6.4 point gap that is invisible to standard fairness metrics because it is a property of the uncertainty rather than of the prediction. The gap is not explained by tighter intervals for women, whose intervals are in fact wider. This is the finding the project regards as its most important, and it was detectable only because coverage was stratified rather than aggregated.

Removing gender from the features does not help, since it is recoverable from proxies at an AUC of 0.654. Post processing mitigation produces an improvement smaller than the split to split noise and is reported as inconclusive. Work experience emerges as the highest leverage intervention while aptitude coaching has no effect.

The contribution is integration and methodological honesty rather than algorithmic novelty. Two practices seem worth generalising beyond this project. Audit the uncertainty and not only the prediction, because a system can pass every standard group fairness test while systematically over-promising to one group. And on datasets of this size, report repeated split results or report nothing, because two conclusions in this project reversed when the evaluation moved from a single split to an average over forty. Every finding here rests on 215 records and is a demonstration of method, not evidence about real cohorts.

6.2 FUTURE SCOPE

The work suggests several extensions, ordered below by the value they would add rather than by the effort they would take.

- **Institutional records with additional protected attributes.** Real, anonymised placement data from this institution, recording category, region, first generation status and disability alongside gender, would lift almost every limitation in Section 6.3 simultaneously. This is the single highest value next step and it is an access problem rather than a technical one.
- **Group conditional conformal calibration.** The coverage gap is a defect this project diagnoses but does not repair. Calibrating the conformity correction separately per group is the standard remedy and should equalise coverage at the cost of wider female intervals. That trade off is worth measuring rather than

assuming.

- **Validation on large real labour market data.** Re-running the full pipeline on American Community Survey data through the folktables interface would give the fairness and coverage statistics genuine statistical weight, with the placement dataset retained as the application demonstration. The original design called for this validation before applying the machinery here, and it was not performed.
- **Oaxaca-Blinder decomposition of the salary gap.** The model reproduces only six percent of the observed salary gap, which suggests a large unexplained component. A formal decomposition reconciled against the SHAP attributions would quantify the explained and unexplained shares directly rather than leaving the inference to a single ratio.
- **Recruiting firm as a feature.** The negative result on salary is attributable to an absent variable rather than to a modelling failure. A dataset that records the recruiting firm as a feature, not only as an outcome, would test that explanation directly and would likely make salary predictable.
- **Multi-attribute and intersectional auditing.** Coverage and group metrics stratified by more than one attribute at a time would reveal whether the over-confidence identified here concentrates in a subgroup rather than distributing evenly across all women.
- **Model card and dataset datasheet.** Publishing the system as a formal model card with an accompanying dataset datasheet would align the artefact with the audit requirements that Local Law 144 and the EU AI Act are establishing, and would make the deployed audit view exportable as a compliance document.
- **Temporal validation across cohorts.** Every result here comes from one institution in one year. Evaluating stability across successive cohorts would establish whether the coverage gap is a property of this sample or of the setting.
- **Counterfactual and multi-feature recourse.** The current recourse search considers single changes only. Extending it to minimal combinations of changes, with a cost model reflecting how hard each change actually is for a student, would help the students for whom the system currently returns no path at all.

6.3 LIMITATIONS

These limitations are material and they are stated without softening, because a report whose central argument is methodological honesty cannot bury its own bounds.

- **Sample size.** 215 students, and the smallest subgroup is 76. Every fairness finding is indicative rather than conclusive. The 6.4 point coverage gap is stable across fifty splits, which is the strongest support available here, but stability across splits is not a significance test.
- **Selection bias.** Salary exists only for the 148 placed students. The salary model describes the placed, not the cohort. The two stage formulation and the if_placed labelling are the honest response, but they scope the claim rather than removing the bias.
- **One protected attribute.** Only gender is recorded. Category, region, first generation status and disability are absent from public data and cannot be audited at all. This is a limitation of the data rather than of the method, and it is a reason to pursue institutional records.
- **Correlation, not causation.** Simulations report what the model believes would follow from a change, not what an intervention would cause. The work experience result in particular is likely confounded, since students who obtain work experience may differ systematically from those who do not.

- **Scope.** One institution, one year, MBA students only. No claim generalises beyond that, and in particular nothing in this report is evidence about the cohorts of the institution at which it was submitted.
- **Unvalidated methods track.** The original design called for validating the conformal and fairness machinery on large real labour market data before applying it here. That validation was not performed, so the methods carry only the guarantees established in their source papers, at a sample size far below what those papers assume.
- **Diagnosed but unrepaired defect.** The coverage gap is identified and quantified but not fixed. A system deployed in this state would be knowingly over-promising to female students, so group conditional calibration is a prerequisite for any real deployment rather than an optional enhancement.
- **No external validity for the recourse advice.** The recourse engine's output has never been tested against what actually happens to a student who follows it. Its value is that it converts a verdict into an actionable statement, not that the statement is verified.

6.4 REPRODUCIBILITY

All code, data references, serialised models and result artefacts are in the repository laid out in Section 3.1. The pipeline is reproduced by running, in order: get_data.py, explore.py, train.py, diagnose_salary.py, fairness.py, explain.py, simulate.py and then tests/smoke_test.py. Every table and figure in this report is generated from reports/*.json by the figure script. The random seed is 42 throughout, coverage results average fifty splits and mitigation results forty.

CHAPTER 7 REFERENCES

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